AWS Solution Architect

Role Overview:
As an AWS Solutions Architect, you design and build scalable, secure, and cost-effective cloud architectures tailored to business needs. Your role involves translating complex requirements into well-architected solutions using AWS services, ensuring high availability, performance, and security. You collaborate closely with stakeholders, developers, and operations teams to guide cloud adoption and optimize infrastructure.

Your responsibilities include:
- Design scalable, secure, and cost-efficient AWS architectures
- Select appropriate AWS services to meet technical and business requirements
- Lead cloud migration and deployment planning
- Create architectural diagrams and documentation
- Ensure compliance with security best practices
- Optimize the performance and cost of cloud infrastructure
- Collaborate with development, operations, and security teams
- Monitor cloud environments and troubleshoot architectural issues
- Provide technical guidance and training to teams
- Participate in capacity planning and disaster recovery design

Q1. A client wants to migrate a multi-tier web application with an on-premises Oracle database to AWS with minimal downtime. How would you plan and design this migration?
Skill Covered: *Cloud migration strategy, database migration, minimal downtime*
Sample Answer:
I'd begin with a thorough assessment of the existing environment, identifying dependencies between web, app, and database tiers. For the database, I'd use AWS Database Migration Service (DMS) with continuous data replication to minimize downtime. The web and app layers would be migrated using a lift-and-shift or re-platforming approach, depending on the application architecture.
To ensure minimal downtime, I'd schedule the cutover during off-peak hours and run parallel systems for validation. Monitoring and rollback plans would be in place. Security groups, VPC setups, and IAM roles would be pre-configured to mirror on-premises access control.

Q2. How would you design an AWS architecture for a globally distributed web application that needs low latency and fault tolerance?
Skill Covered: *Global architecture, fault tolerance, content delivery*
Sample Answer:
I'd deploy the application in multiple AWS Regions using Route 53 with latency-based routing to direct users to the nearest healthy region. For static content, Amazon CloudFront CDN would cache assets at edge locations worldwide, reducing latency. Data replication would be handled with multi-region databases or DynamoDB Global Tables. Failover is ensured by health checks and Route 53 DNS failover, while auto-scaling groups maintain capacity. Multi-AZ deployments inside regions improve fault tolerance.

Q3. A customer's AWS bill has unexpectedly spiked. What tools and approaches would you use to identify and resolve the issue?
Skill Covered: *Cost optimization, billing analysis, AWS tools*
Sample Answer:
I'd start by analyzing Cost Explorer and AWS Budgets to pinpoint the services driving the cost increase and detect anomalies. Trusted Advisor provides recommendations on idle resources, underutilized instances, or unassociated Elastic IPs.
After identifying waste, I'd implement rightsizing of instances, terminate unused resources, and recommend Reserved Instances or Spot Instances for predictable workloads. Setting up cost alerts helps prevent future surprises.

Q4.Explain how you'd secure an S3 bucket containing sensitive financial data, including access and encryption controls.
Skill Covered: *Security best practices, encryption, access management*

Sample Answer:
I'd enforce S3 Block Public Access settings to prevent accidental exposure. Access would be tightly controlled with IAM policies and bucket policies granting least privilege. Multi-factor authentication could be enabled for critical access.Encryption would be enabled using SSE-KMS to manage keys securely, and data in transit would be protected with SSL/TLS. Versioning would be activated to protect against accidental deletion, and CloudTrail logging would monitor access for auditing.

Q5.Describe your approach to designing a disaster recovery solution with RTO under 1hour for a critical application.
Skill Covered: *Disaster recovery planning, backup strategies, RTO/RPO*
Sample Answer:
I'd use Multi-AZ deployments for high availability within a region and enable cross-region replication for critical data. Backups using EBS snapshots and RDS automated backups would be scheduled frequently.
Route 53 health checks combined with DNS failover ensure traffic rerouting during outages.
Infrastructure is codified with CloudFormation for quick environment rebuild. Regular DR tests validate readiness and identify gaps.

Q6. How do you ensure that your AWS environment complies with organizational security policies and industry standards?
Skill Covered: *Compliance, monitoring, governance*
Sample Answer:
I implement AWS Config rules to continuously check resource compliance against organizational policies like encrypted storage and restricted public access. AWS Security Hub aggregates security findings for centralized management.CloudTrail captures all API activity for audit purposes. Automated remediation via Lambda functions and regular audits ensures compliance. Documentation and training promote best practices.

Q7. A new microservices-based app will be deployed on AWS. How would you design the networking and security architecture?
Skill Covered: *Microservices architecture, networking, security groups*
Sample Answer:
I'd deploy services in private subnets within a VPC and control access using Security Groups and Network ACLs. Service-to-service communication would use tightly scoped security group rules and possibly a service mesh for observability. API Gateway would handle external access with throttling and authentication (e.g., Cognito). Data services like databases would be isolated in private subnets. Encryption in transit and at rest would be enforced throughout.

Q8. How do you approach designing for scalability when workload patterns are unpredictable?
Skill Covered: *Auto-scaling, elasticity, cost management*
Sample Answer:
I'd use Auto Scaling Groups with dynamic scaling policies tied to CloudWatch metrics like CPU or request latency. Load balancers distribute incoming traffic to healthy instances. To control costs, I'd mix On-Demand and Spot Instances and consider serverless options like Lambda where appropriate. Monitoring usage trends enables proactive capacity planning.

Q9. What AWS services and design patterns would you use to build a real-time data processing pipeline?
Skill Covered: *Data streaming, event-driven architecture*
Sample Answer:
I'd use Kinesis Data Streams or Amazon Managed Streaming for Kafka (MSK) to ingest real-time data. Lambda functions or AWS Glue would process the data, with results stored in DynamoDB or S3 for analysis. Event-driven patterns ensure scalability and decoupling. CloudWatch monitors pipeline health, and AWS IAM secures data access.

Q10. A legacy application uses a single database and cannot be refactored easily. How would you migrate it to AWS while improving resilience?
Skill Covered: *Lift-and-shift migration, resilience, RDS*
Sample Answer:
I'd perform a lift-and-shift migration by moving the database to Amazon RDS with Multi-AZ enabled for failover. The application is migrated as-is to EC2 instances within an Auto Scaling Group across AZs.

I'd implement backups, monitoring, and security best practices. Although the app isn't refactored, AWS services ensure higher availability and easier maintenance.

AWS DevOps Engineer

Role Overview:
As an AWS DevOps Engineer, you automate infrastructure provisioning, build scalable CI/CD pipelines, and maintain cloud infrastructure to enable rapid, reliable software delivery. You collaborate with development and operations teams to streamline deployment, monitor applications, and ensure security and compliance in AWS environments.

Your responsibilities include:
- Develop and manage CI/CD pipelines using AWS services (CodePipeline, CodeBuild, CodeDeploy)
- Automate infrastructure provisioning and configuration with CloudFormation, Terraform, or AWS CLI
- Monitor application and infrastructure health using CloudWatch and set automated alerts
- Implement security best practices in deployment pipelines and infrastructure
- Manage container orchestration with ECS, EKS, or Kubernetes on AWS
- Collaborate with developers to optimize build, test, and deployment workflows
- Troubleshoot and resolve issues in the CI/CD pipeline and cloud infrastructure
- Maintain documentation of automation scripts, deployment processes, and troubleshooting steps

Q1. How do you implement a CI/CD pipeline on AWS for a microservices application?
Skill Covered: *CI/CD pipeline design, AWS DevOps tools integration*
Sample Answer:
I use a GitHub repository as the source code repository, which triggers AWS CodePipeline upon code commits. CodeBuild handles the build and test stages for each microservice, ensuring isolated and parallel builds. CodeDeploy automates deployments to ECS or EKS clusters, managing containerized microservices.
I incorporate automated tests and manual approval stages where necessary. Monitoring with CloudWatch Events and SNS alerts enables proactive pipeline health management, ensuring reliable and fast delivery.

Q2. Explain Infrastructure as Code and how you've used it with AWS.
Skill Covered: *Infrastructure automation, CloudFormation*
Sample Answer:
Infrastructure as Code (IaC) involves defining cloud resources declaratively or programmatically to automate provisioning. I primarily use AWS CloudFormation templates to create and update stacks, ensuring environments are consistent and repeatable. For multi-cloud or complex scenarios, I use Terraform to manage AWS resources. IaC enables version control, reduces manual errors, and accelerates infrastructure deployments, critical for agile development.

Q3. How do you monitor and troubleshoot failed deployments in a CI/CD pipeline?
Skill Covered: *Monitoring, troubleshooting, CloudWatch, logging*
Sample Answer:
I configure CloudWatch Logs and Events to capture deployment logs and pipeline execution states. When a deployment fails, I examine CodeBuild and CodeDeploy logs to identify errors such as build failures, permission issues, or resource conflicts.
I use CloudWatch Alarms to alert the team immediately. Root cause analysis involves verifying IAM roles, network connectivity, and service limits, followed by fixing configuration or code errors to restore pipeline health.

Q4. What strategies do you use to secure the CI/CD pipeline?
Skill Covered: *Security automation, IAM, secrets management*
Sample Answer:
I implement least privilege IAM roles for pipeline components and encrypt artifacts at rest and in transit. Sensitive data like API keys and credentials are stored securely using AWS Secrets Manager or Systems Manager Parameter Store.
Pipeline stages include static code analysis and vulnerability scans. Manual approval gates prevent unauthorized deployments to production, enhancing security without hindering agility.

Q5. Describe how you manage container orchestration on AWS.
Skill Covered: *Container management, ECS, EKS, Kubernetes*
Sample Answer:
I manage containers primarily with Amazon ECS for simpler setups or EKS for Kubernetes-managed orchestration. ECS integrates seamlessly with other AWS services and simplifies deployment with task definitions and service discovery.
EKS provides full Kubernetes compatibility, ideal for complex microservices needing advanced scheduling and scaling. I use Fargate for serverless container execution, minimizing infrastructure management.

Q6. How do you implement blue/green deployments using AWS services?
Skill Covered: *Deployment strategies, AWS CodeDeploy, traffic shifting*
Sample Answer:
Using AWS CodeDeploy, I set up two identical environments (blue and green). New versions are deployed to the green environment while the blue serves live traffic. Once verified, traffic is shifted gradually to green using weighted routing with ELB or Route 53. This strategy reduces downtime and enables quick rollback by shifting traffic back if issues arise.

Q7. How do you automate configuration management and application deployments?
Skill Covered: *Automation, AWS Systems Manager, CodeDeploy*
Sample Answer:
I use AWS Systems Manager for patching, configuration updates, and running commands across EC2 instances. CodeDeploy automates application deployments with lifecycle hooks for pre- and post-deployment scripts.Combining these tools allows consistent and repeatable deployments with minimal manual intervention.

Q8.What monitoring metrics do you track to ensure the health of a deployed application?
Skill Covered: *Monitoring, CloudWatch metrics, proactive incident management*
Sample Answer:
Key metrics include CPU utilization, memory usage, disk I/O, network latency, and error rates.
For containers, I monitor task health and scaling events. Application-specific logs and custom metrics provide deeper insights.
CloudWatch Alarms notify when thresholds exceed safe limits, triggering automated remediation or alerts to the team.

Q9. How have you handled rollbacks in your deployment pipelines?
Skill Covered: *Deployment rollback, disaster recovery, pipeline management*
Sample Answer:
I implement rollback strategies like automatic rollback on deployment failure within CodeDeploy, or blue/green deployments to switch traffic between stable and new versions.
Rollbacks are tested regularly to ensure they restore application availability quickly.
Documentation and versioning support quick identification and deployment of previous stable Releases.

Q10.How do you integrate security scans into your CI/CD pipeline?
Skill Covered: *DevSecOps, security automation, vulnerability scanning*
Sample Answer:
I integrate static code analysis tools (e.g., SonarQube) and container vulnerability scanners (e.g., Amazon ECR image scanning) into CodeBuild stages. Failures on critical vulnerabilities halt the pipeline.
Regular updates of scanning tools and policies ensure security remains current. This proactive approach embeds security throughout the development lifecycle

AWS Security Engineer

Role Overview:
As an AWS Security Engineer, you design, implement, and maintain security controls and policies to protect AWS environments. Your role involves risk assessment, compliance enforcement, monitoring for threats, and incident response. You work closely with infrastructure and development teams to embed security best practices throughout the cloud lifecycle, ensuring confidentiality, integrity, and availability of data and applications.

Your responsibilities include:

- Develop and enforce AWS security policies, controls, and governance frameworks
- Manage IAM roles, policies, and access management using least privilege principles
- Design and implement network security controls (VPC, Security Groups, NACLs)
- Monitor AWS environments using CloudTrail, GuardDuty, Security Hub, and CloudWatch
- Conduct vulnerability assessments, penetration testing, and security audits
- Implement encryption solutions for data at rest and in transit
- Respond to and investigate security incidents and breaches
- Provide security training and awareness to teams and stakeholders

Q1. How do you secure AWS IAM to prevent unauthorized access?
Skill Covered: Identity and access management, security best practices
Sample Answer:
I enforce the principle of least privilege by creating IAM policies that grant only the necessary permissions to users and roles. I enable multi-factor authentication (MFA) for all privileged accounts to add an extra layer of security. Root account usage is strictly limited and monitored. Additionally, I regularly review IAM roles and policies to remove unused permissions and rotate access keys periodically. Using IAM Access Analyzer helps detect any unintended resource sharing.

Q2. What steps would you take to protect data stored in Amazon S3?
Skill Covered: *Data encryption, access control, S3 security features*
Sample Answer:
I block all public access by enabling S3 Block Public Access settings and configure bucket policies to restrict access to authorized IAM users or roles. Server-side encryption with AWS KMS (SSE-KMS) ensures data at rest is encrypted and provides key management control. For data in transit, I enforce HTTPS connections. I enable versioning and logging to track data changes and access patterns, which helps in auditing and recovering from accidental deletions or overwrites.

Q3. How does AWS GuardDuty work and how would you respond to a threat detected by it?
Skill Covered: Threat detection, monitoring, and incident response**Sample Answer:**
GuardDuty continuously monitors AWS accounts and workloads by analyzing CloudTrail logs, VPC flow logs, and DNS logs for suspicious activities like unauthorized access or reconnaissance. When GuardDuty detects a threat, it generates security findings with severity levels.
Upon receiving an alert, I investigate the finding details, isolate affected resources if needed, revoke compromised credentials, and apply remediation steps such as updating security groups or patching vulnerabilities. I also document the incident and improve preventive controls.

Q4. Explain how you would implement network security in AWS.
Skill Covered: Network segmentation, firewall rules, VPC security
Sample Answer:
I design VPCs with private and public subnets to segregate sensitive workloads. Security groups act as virtual firewalls for instances, controlling inbound and outbound traffic based on IPs, ports, and protocols. Network ACLs provide an additional layer of subnet-level stateless filtering.
I also configure VPN or Direct Connect for secure hybrid connectivity and use AWS WAF to protect web applications from common attacks. Regular audits ensure network policies align with security standards.

Q5. What are the key AWS services you use for logging and auditing?
Skill Covered: *Logging, auditing, and compliance*
Sample Answer:
AWS CloudTrail captures all API calls and user activity for auditing and forensic investigations.Amazon CloudWatch collects logs and metrics for performance and operational monitoring.
AWS Config continuously assesses resource configurations against compliance rules.
For centralized security findings, AWS Security Hub aggregates data from GuardDuty, Inspector, and other tools, helping to maintain an overall security posture.

Q6. How do you encrypt data in transit and at rest in AWS?

Skill Covered: Data encryption, key management
Sample Answer:
For data at rest, I use AWS KMS-managed keys to enable encryption on services like S3, EBS, and RDS. Server-side encryption with KMS (SSE-KMS) provides fine-grained access control and audit logs.
Data in transit is protected using TLS/SSL protocols. For internal communication within VPCs, I implement mutual TLS where applicable and enforce encryption for VPN tunnels.

Q7. Describe a time when you detected and responded to a security incident in AWS.
Skill Covered: *Incident detection, response, and remediation*
Sample Answer:
In one instance, GuardDuty alerted us to unusual API calls from an unfamiliar IP. I immediately isolated the compromised IAM user, revoked its credentials, and conducted a root cause analysis using CloudTrail logs.
We patched the vulnerability exploited and updated IAM policies to restrict permissions further.
A post-incident review led to improved monitoring and automated response workflows to prevent recurrence.

1. What is _Amazon Web Services (AWS)_ and what are its key features?

Amazon Web Services (AWS) is a cloud computing platform that offers a vast range of services, including computing power, storage, and databases, to help businesses scale and grow more cost-effectively.

Key Features of AWS

1. Scalability and Elasticity: AWS provides tools that allow for both vertical and horizontal scaling, as well as the ability to auto-scale based on demand.
2. Global Reach: With data centers in multiple geographic regions, AWS enables businesses to operate on a global scale while remaining compliant with local regulations.
3. Pay-As-You-Go Pricing: This flexible pricing model allows users to pay only for the resources they consume, reducing upfront costs.
4. Security and Compliance: AWS offers a variety of security tools and features to help protect data, as well as compliance with numerous industry standards.
5. Hybrid Capabilities: AWS supports hybrid architectures, allowing businesses to integrate their existing on-premises solutions with the cloud.
6. Artificial Intelligence and Machine Learning: With AWS, businesses can harness the power of AI and ML through accessible services for data processing, analysis, and more.
7. Developer Tools: From code repository management to continuous integration and deployment, AWS provides a comprehensive suite of developer-centric services.
8. Internet of Things (IoT): AWS offers capabilities for managing and processing IoT data, connecting devices securely to the cloud.

2. Explain the concept of _Regions_ and _Availability Zones_ in AWS.

When working with AWS, it's important to understand the fundamental building blocks of Regions and Availability Zones.

AWS Regions
An AWS Region is a separate geographic area, often a specific city or part of a country, with multiple, distinct data centers. Each Region is designed to be largely self-contained and is connected to other Regions through high-speed, secure networks.

AWS Availability Zones
An AWS Availability Zone (AZ) is a distinct, separate building or data center within a Region. These AZs are interconnected with high-bandwidth, low-latency networking, enabling redundancy and fault tolerance.

Key Characteristics

- Isolation: Each AWS Region is completely isolated from other Regions in terms of its infrastructure, and is designed to be a standalone unit.
- Number of AZs: Most AWS Regions are composed of at least three AZs, although some may have more. The use of three or more AZs is aimed at providing a comprehensive high-availability solution.
- Distance: The AZs within a Region are located in close geographical proximity to each other, typically within 100 miles to ensure low latency.

Benefits of Using Regions and AZs
- High Availability: Deploying resources across multiple AZs within the same Region ensures redundancy and high availability.
- Fault Tolerance: AZs are designed to be isolated from one another in terms of most failure scenarios, providing a level of fault tolerance that can safeguard against localized outages.

Considerations for Multi-Region Architectures
- Latency: When designing multi-Region architectures, latency due to geographic distances between Regions must be taken into account.
- Data Replication: Multi-Region setups often require robust data replication strategies to ensure data consistency and integrity.

3. What is an _Amazon Machine Image (AMI)_ and how is it used?
An Amazon Machine Image (AMI) serves as a template for virtual servers, known as EC2 instances, enabling rapid and consistent provisioning.

Core Components
- Launch Permissions: Dictate which AWS accounts can utilize the AMI to instantiate EC2 instances.
- Block Device Mapping (BDM): Prescribes the storage volumes, such as EBS snapshots and their characteristics, that should be attached to the launched instance.

Unique AMI Features
- Root Volume: The AMI captures a state, including the operating system and pre-installed applications, and stores it as a template. This template is employed to establish the root volume of the EC2 instance.
- Customizations and Snapshots: Beyond the root volume, the AMI can include additional storage volumes. These might have specific data sets or applications. When the AMI is utilized to launch an EC2 instance, it ensures that these configured volumes are also established and linked to the new EC2 instance.

Benefits and Use Cases
- Efficiency: With AMIs, it's feasible to create highly-tailored EC2 instance configurations that launch swiftly. This characteristic is beneficial for auto-scaling groups.
- Provisioning Consistency: Teams can guarantee that every EC2 instance, whether for development, testing, or production, commences with an identical setup, as defined by the AMI.
- Operational Safety: AMIs serve as backups. In adverse situations, such as data corruption, a previous AMI can be utilized to restore a functional EC2 instance.

AMI IDs in Code
Here is a Python boto3 script example:

```python
import boto3

# Establish a connection to the EC2 service
ec2 = boto3.client('ec2')

# Retrieve details of all owned AMIs
owned_amis = ec2.describe_images(Owners=['self'])

# Display the ID of each owned AMI
```

```
for ami in owned_amis['Images']:
    print(ami['ImageId'])
```

4. Describe the difference between _Elastic IP_ and _Public IP_ in AWS.
In AWS, every EC2 instance automatically gets a Public IP and can optionally be assigned an Elastic IP for more flexibility.

Public IP
- Dynamic: Assigned when the instance starts and lost on stop or termination.
- Shared: Drawn from a pool of AWS addresses, potentially used by other instances.
- Cost: Free while associated with a running instance.

Useful for instances that need temporary, internet-facing access.

Elastic IP
- Static: Remains constant until explicitly released.
- Dedicated: Solely assigned to the AWS account unless released.
- Cost: Incurs charges when not in use with a running instance.

Designed for hosting applications or network appliances that require a consistent public IP address.

Best Practices
- Public IP: Let instances use public IPs unless there's a specific need for a static address. Avoid leaving unused Elastic IPs assigned to instances, as this costs money. Instead, consider releasing them and using other appropriate mechanisms, such as public IPs or AWS resources like load balancers and NAT gateways.

5. What is the role of the _AWS Management Console_?
The AWS Management Console serves as a graphical user interface for interacting with AWS services. It offers an intuitive way to access and manage a wide range of AWS resources.

Key Features
- User-Friendly Dashboard: Provides an overview of system health, cost management tools, and recent resource activity.
- Service Navigation: Offers categorized service access, from compute and storage to machine learning and analytics.
- Resource Management: Allows for resource provisioning, configuration, and monitoring through a point-and-click interface.
- Task Automation: Enables set-up of automated tasks such as backup schedules and resource scaling.
- Integrated Tools: Incorporates third-party applications and AWS-specific utilities for enhanced functionality.
- Collaboration and Security: Facilitates user and access management, making it easier to work within teams while adhering to best security practices.

Console vs. SDKs vs. CLIs
Compared to the AWS Command Line Interface (CLI) and Software Development Kits (SDKs) for various programming languages:

 # Key Distinctions
- Ease of Use: The console's graphical nature makes it more approachable for beginners, whereas CLIs and SDKs cater more to developers and advanced users.
- Functionality: The console covers a broad range of AWS services, but might lag behind in supporting the latest offerings compared to the up-to-date coverage provided by SDKs.
- Workflow Flexibility: CLIs are often preferred for scripting and automation, while the console is tailored for manual, point-and-click operations.

Console Role-Based Access
When the AWS Management Console is used in conjunction with the Identity and Access Management (IAM) service, it allows for granular, role-based controls and shared access across teams.

6. Explain the concept of _Elastic Computing_ in AWS.
Elastic Computing in AWS refers to the ability to dynamically adjust your computing needs based on real-time demands and pay only for what you use. This is achieved through services like Amazon EC2.

Key Features
- Scalability: EC2 instances can be scaled up or down to accommodate varying workloads.
- Load Balancing: Multiple EC2 instances can distribute incoming traffic for improved performance.
- Auto Scaling Groups: EC2 instances can be automatically scaled in response to changing demand patterns.
- Application Load Balancers: Tailored for handling HTTP and HTTPS traffic.

Benefits
- Cost-Efficiency: Pay-as-you-go model reduces expenses related to underutilized resources.
- Performance: Elasticity ensures that sufficient resources are available to meet performance requirements.
- Fault Tolerance: Using multiple EC2 instances and Auto Scaling enhances system reliability.

Code Example: Auto Scaling Setup
Here is the Python code:

```python
import boto3

client = boto3.client('autoscaling')

response = client.create_auto_scaling_group(
    AutoScalingGroupName='string',
    LaunchConfigurationName='string',
    MinSize=1,
    MaxSize=1,
    DesiredCapacity=1
)
```

7. What is _AWS Identity and Access Management (IAM)_ and why is it important?
AWS Identity and Access Management (IAM) is a free AWS service that grants secure access to AWS resources. It enables you to control who can use your AWS resources (authentication) and how they can use them (authorization).

Key Components
1. Users: These are the end users who would be accessing the AWS resources. They can be grouped together according to the designations or roles.
2. Groups: Groups are a way to combine several users so that they can be assigned the same set of permissions. This makes managing permissions easier, especially in scenarios where multiple users require similar levels of access.
3. Roles: IAM roles are created and then assigned to other AWS resources or AWS accounts. They eliminate the need to share long-term credentials. Instead, they allow for secure access to resources.

Why IAM is important
IAM is fundamental to AWS security and offers several advantages:

1. Principle of Least Privilege: Ensures users and resources have only the permissions they need to perform their tasks, reducing risks.
2. Granular Permissions: AWS provides a vast range of services, and within each service, there are numerous actions. IAM allows for specific actions on particular services to be granted, offering a great degree of control.
3. Access Management to Resources: IAM not only manages access for users and groups but also for services, ensuring secure communication between AWS resources.

4. Secure Access Sharing: Using roles, AWS allows for secure cross-account sharing. This is used by organizations that have multiple AWS accounts to enforce security and centralize management.
5. Compliance Tracking: IAM provides detailed logs to track user activity, which is crucial for compliance with industry standards.
6. Password Policies: IAM allows for strong password policies, ensuring user authentication methods comply with security best practices.

8. Describe the _AWS Shared Responsibility Model_.
The AWS Shared Responsibility Model establishes clear responsibilities for security and compliance between AWS and the customer. This model varies for different AWS services, but generally follows two core components: "Security of the Cloud" and "Security in the Cloud".

Security of the Cloud
AWS holds the primary responsibility for protecting the infrastructure and physical facilities on which its services are built. This includes:
- Global Infrastructure: AWS maintains secure data centers, with measures such as biometric access control and continuous surveillance.
- Compliance Certifications: AWS obtains third-party security and compliance certifications to ensure its operations meet industry-standards.
- Hardware and Software: AWS manages the security and maintenance of the hardware and software infrastructure that powers its cloud services.

Security in the Cloud
Customers are responsible for securing their data, applications, and services running on AWS. The extent of this responsibility can vary based on the specific AWS service in use, but generally includes:
- Data Encryption: Customers should encrypt their data to protect its confidentiality and integrity during transit and while at rest in AWS services.
- Access Management: Implementing robust Identity and Access Management (IAM) policies to regulate user access to AWS resources.
- Operating System and Networking: For Infrastructure as a Service (IaaS) offerings, customers are responsible for securing their Operating Systems and network configurations, among other tasks.
- Configuration Management: Customers should manage and monitor the configuration of their AWS resources to ensure they align with best security practices.

9. What is the difference between _Vertical Scaling_ and _Horizontal Scaling_ in AWS?
Vertical scaling involves increasing the resources of a single machine, such as its CPU or RAM. In contrast, horizontal scaling means adding more machines to a network, distributing the workload across them.

10. Explain the concept of _High Availability_ in AWS.
High Availability (HA) in AWS ensures that your applications and data are accessible and resilient to hardware and software failures. AWS achieves high availability through a combination of fault-tolerant design, redundancy, and automated recovery mechanisms.

Core Components for High Availability
1. Availability Zones (AZs): These are isolated data centers within a geographic region. Using multiple AZs helps in achieving fault isolation.
2. Auto Scaling Groups: These dynamically manage the number of EC2 instances based on real-time demand.
3. Elastic Load Balancing (ELB): Distributes incoming traffic across multiple EC2 instances to ensure balanced load and immediate failover in case of issues.
4. Amazon CloudWatch: Monitors your AWS resources and the applications you run on AWS.
5. Amazon Route 53: Provides reliable and cost-effective domain registration with built-in DNS routing.

 # Code Example: Auto Scaling Group
Here is the Python code:

```python
import boto3

client = boto3.client('autoscaling')

response = client.create_auto_scaling_group(
    AutoScalingGroupName='string',
    LaunchConfigurationName='string',
    MinSize=1,
    MaxSize=3,
    DesiredCapacity=2,
    AvailabilityZones=[
        'string',
    ],
    LoadBalancerNames=[
        'string',
    ]
)
```

 # Code Example: Elastic Load Balancer
Here is the Python code:

```python
import boto3

client = boto3.client('elbv2')

response = client.create_load_balancer(
    Name='MyLoadBalancer',
    Subnets=[
        'subnet-0e541b6eb61bb736c',
    ],
    SecurityGroups=[
        'sg-04bbe83913172e35e',
    ],
    Type='application'
)
```

AWS Services for High Availability
- Compute: AWS provides services like EC2, ECS, EKS, and Lambda for high availability of your compute resources.
- Storage: Services such as S3 for object storage, EBS for block storage, and EFS for file storage ensure high availability of your data.
- Databases: AWS RDS, DynamoDB, and Redshift are built to provide highly available database solutions.
- Networking: AWS Direct Connect, VPC, and VPN ensure a highly available network architecture.

11. What is _Amazon Elastic Compute Cloud (EC2)_ and what are its key features?
Amazon Elastic Compute Cloud (EC2) is a web service that provides resizable compute capacity in the cloud. It is designed for developers to have full control over computing resources in a highly available and cost-effective manner.

Key Features

1. Virtual Computing Environment: EC2 enables users to set up virtual machines, known as instances, for running their applications. These instances function like real computers and are hosted in the cloud.
2. Variety of Instance Types: EC2 offers diverse instance families optimized for various workloads, such as general-purpose computing, memory or CPU-intensive tasks, storage-optimized applications, and more.
3. Purchasing Options: Users can select from on-demand instances (pay-as-you-go), spot instances (bid for unused capacity at potentially lower costs), and reserved instances (long-term contracts for reduced pricing).
4. Integrated Security: Security Group and Virtual Private Cloud (VPC) mechanisms help in controlling network access to instances, and Key Pairs facilitate secure instance logins.
5. Scalability and Elasticity: EC2 supports auto-scaling to adjust instance capacity based on demand, and Elastic Load Balancing to distribute traffic across multiple instances.
6. Custom AMIs: Users can create customized Amazon Machine Images (AMIs) to encapsulate specific software configurations and resources.
7. Flexible Storage Options: Amazon EC2 provides various types of storage volumes, including Amazon EBS for persistent block storage and Amazon S3 for object storage.
8. Network Performance Monitoring: Users can monitor the network performance of their instances with tools like Elastic Network Adapters (ENAs) and Enhanced Networking.
9. Integrated Ecosystem: AWS Management Console, AWS Command Line Interface (CLI), and Software Development Kits (SDKs) streamline EC2 instance management.
10. Resource Tagging: Tags help in managing and organizing resources by providing metadata for instances.

12. Describe the different _EC2 instance types_ and their use cases.
Amazon EC2 offers a broad range of instance types optimized to fit different use cases. These types can be categorized into groups like General Purpose, Compute Optimized, Memory Optimized, Storage Optimized, and Accelerated Computing.

General Purpose
Use Cases
These instance types are suitable for a diverse array of workloads, from small to medium databases to development and testing environments.

Burstable Performance
- T2: Designed for cost-efficient applications with short bursts of CPU usage. Accumulates CPU credits during low usage, which can then be used during traffic spikes.

Compute Optimized
Use Cases
Ideal for compute-bound applications requiring high performance from the CPU.
- C5: Utilizes high-frequency Intel Xeon Scalable processors.
- C6g and C6gn: Powered by AWS Graviton2 processors, which are based on Arm architecture, and provide the best price-performance in the compute-optimized category.

 # Code Example: c5 Instance
Here is the code:

```plaintext
c5.large
vCPU: 2
RAM: 4 GB
Networking: Up to 10 Gbps
Storage: EBS-Only
Price: Moderate
```

 # Code Example: t2 Instance
Here is the code:

```plaintext
t2.micro
vCPU: 1
RAM: 1 GB
Networking: Low to Moderate
Storage: EBS-Only
Price: Low
```

Memory Optimized
Use Cases
Suited for memory-intensive applications like high-performance databases, distributed memory caches, and in-memory analytics.
- X1e: Offers the most memory in a single instance.
- R6g and R6gn: Utilizes AWS Graviton2 processors and provides a balance of compute, memory, and networking resources at a lower cost.

Storage Optimized
Use Cases
Designed for applications demanding high, sequential read and write access to very large data sets, like data warehousing and Hadoop clusters.
- I3: Utilizes Non-Volatile Memory Express (NVMe)-based SSDs for extremely high random I/O performance.
- D2: Cost-effective option for workloads that require high sequential read/write performance.

Accelerated Computing
Use Cases
Ideal for compute-intensive workloads that can benefit from the parallel processing capabilities of GPUs.
- P3: Equipped with NVIDIA Tesla V100 GPUs, suitable for deep learning, computational fluid dynamics, and computational finance.
- G4dn: Combines NVIDIA T4 GPUs with custom Intel Cascade Lake CPUs, optimized for gaming, machine learning, and 3D visualization.
- F1 and A1: Designed for specific workloads using FPGAs (Field-Programmable Gate Arrays) and AWS Inferentia, respectively.

13. What is _Amazon Elastic Container Service (ECS)_ and how does it work?
Amazon Elastic Container Service (ECS) is a highly scalable, high-performance container management service that supports Docker containers and allows you to easily run applications on Amazon EC2 and AWS Fargate.

Key Features
- AWS Fargate Integration: Run containers without provisioning or managing servers.
- Task and Service Definitions: Define and configure your tasks and services using the ECS management console or task definitions.
- Service Auto Scaling: Automatically adjust service capacity based on load.
- Service Load Balancing: Balance incoming traffic across containers in a service.
- Task Scheduling: Place tasks based on resource needs, strategies, and state.
- Custom Schedulers: Integrate third-party or custom schedulers for advanced orchestration.

ECS Core Concepts

 # Cluster
A logical grouping of tasks and services. It acts as a base to host tasks and services. Within a cluster, you can have both EC2 instances and/or AWS Fargate capacity to run tasks.

 # Task Definition

This is where you specify what container images to use, and various container settings like networking and storage. Think of a task definition as a blueprint for your application.

Task
An instantiation of a task definition that's running on the cluster.

Service
Ensures that a specified number of tasks from a task definition are running and available. If any tasks or instances fail or are terminated, the service automatically launches new instances to maintain the desired number of tasks.

Container Agent
For ECS to function, your EC2 instances must have the ECS container agent running on them. This agent communicates with the ECS service in AWS, allowing tasks to be launched on the instance.

ECS Modes
ECS on EC2
- Instance Management Responsibility: You're responsible for provisioning and managing EC2 instances in your cluster.

AWS Fargate
- Serverless: Run containers without managing the underlying infrastructure.
- Task Level Responsibility: You define tasks and their requirements; AWS handles the rest.

ECS Pricing
ECS pricing follows a pay-as-you-go model, where you're charged based on the AWS resources you use with ECS. There are costs associated with networking, storage, EC2 or Fargate usage, as well as any AWS integrations like load balancing or CloudWatch.

14. Explain the difference between _Amazon EC2_ and _AWS Lambda_.
Amazon EC2 (Elastic Compute Cloud) and AWS Lambda offer compute services, but they differ in their paradigms of use.

Amazon EC2
- Virtual Servers: EC2 provisions virtual machines, giving you full control over the operating system.
- Instance Types: Offers a wide range of instance types optimized for various workloads, such as compute-optimized, memory-optimized, and storage-optimized.
- Pricing Model: Uses a pay-as-you-go model, with pricing based on the type and size of the instance, as well as any additional resources used (e.g., storage, data transfer).
- Use Case Flexibility: Ideal for predictable workloads or applications that require long-running, consistent compute resources.

AWS Lambda
- Serverless Compute: Lambda runs code in response to specific events and automatically scales based on the incoming workload, without requiring you to manage underlying servers.
- Stateless Execution: Each function invocation is independent, without any persistent state between invocations.
- Event-Driven: Designed for workloads that are triggered by AWS services or HTTP requests.
- Cost Efficiency: Billed based on the number of executions and the compute time used, making it cost-effective for sporadic workloads.
- Programming Languages: Offers broader language support with the freedom to run custom code.

 # Key Distinctions
- Resource Management: EC2 requires you to manage and monitor your instances, while Lambda abstracts infrastructure management.
- Startup Latency: EC2 instances are pre-provisioned, offering immediate compute resources. Lambda, while highly scalable, might experience slight startup delays as it initializes resources based on the incoming workload.
- Operating Models: EC2 aligns with a more traditional virtual server model, while Lambda embodies the serverless, event-driven paradigm.

- Compute Duration: EC2 gives you full control over how long you want to keep an instance running, while Lambda functions have a maximum execution duration (default of 15 minutes).
- Scalability: Both EC2 and Lambda are designed to scale based on demand, but Lambda provides more automated scaling based on the number of incoming events.

15. What is _AWS Elastic Beanstalk_ and when would you use it?
AWS Elastic Beanstalk is a Platform as a Service (PaaS) that streamlines the deployment and management of cloud-based applications. It automatically handles infrastructure provisioning, load balancing, auto-scaling, and more, allowing developers to focus primarily on writing code.

Key Features
- Application Management: Elastic Beanstalk supports various application types, including Docker, Go, Java, .NET, Node.js, PHP, Python, and Ruby. It also caters to both web applications and services via its web interface and HTTP API support.
- Configurational Flexibility: Users can opt for simple, predefined configurations or exercise fine-grained control over resources for advanced setups.
- Deployment Options: Beanstalk accommodates multiple deployment methods, such as from a Git repository, using the EB Command Line Interface (CLI), or through the AWS Management Console.
- Monitoring and Logging: The service integrates with Amazon CloudWatch for monitoring and provides options for enhanced logging.

When to Use Elastic Beanstalk
Beanstalk is especially beneficial for:
- Rapid Deployment: It empowers quick deployment without the need for in-depth AWS knowledge.
- Resource Optimization: Automated provisioning decreases the likelihood of over- or under-provisioning resources.
- Focus on Development: It suits situations where developers prefer a managed environment, freeing them from infrastructural concerns.
- Cost Efficiency for Development: It can help keep development costs in check, but might not be the most cost-efficient for large-scale, long-running applications due to its pricing model's lack of granular control.

Code Example: Basic Elastic Beanstalk Configuration
Here is the Python script:

```python
from flask import Flask
application = Flask(__name__)

@application.route("/")
def hello():
    return "Hello World!"

if __name__ == "__main__":
    application.run()
```

Here is the configuration file for Elastic Beanstalk deployment:
```yaml
option_settings:
  aws:elasticbeanstalk:environment:process:default:
    Timeout: '20'
  aws:autoscaling:launchconfiguration:
    InstanceType: t2.micro
resources: {}
```

1. Core Application Deployment (REST API/Compute)
EC2 (Elastic Compute Cloud): This is AWS's compute service for running virtual machines (VMs).
Elastic Load Balancer (ELB): Used to distribute incoming application traffic across multiple EC2 instances to ensure high availability and fault tolerance.
Autoscaling Group (ASG): Enables automatic scaling of EC2 instances based on demand, ensuring applications can handle varying loads.
Regions and Zones: AWS uses these concepts to ensure high availability and disaster recovery by geographically distributing VMs across different data centers.

2. Managed Compute Services
To simplify deployment and management, AWS offers managed services:
Elastic Beanstalk: A managed service for deploying and scaling web applications and services without worrying about the underlying infrastructure. It supports various languages and container images.
Container Orchestration: For microservices architectures, AWS offers:
Elastic Kubernetes Service (EKS): A managed Kubernetes service for running containerized applications.
Elastic Container Service (ECS): AWS's own container orchestration service.
Fargate: A serverless compute engine for containers, allowing users to run containers without managing servers or clusters.
Serverless Computing with AWS Lambda: This service allows you to run code without provisioning or managing servers, paying only for the compute time consumed.

3. Database Services
The video then shifts to database options in AWS, differentiating between SQL and NoSQL databases:
Relational Databases (SQL):
Amazon RDS (Relational Database Service): A managed service supporting various relational database engines like MySQL, PostgreSQL, Oracle, and SQL Server.
Aurora: A MySQL and PostgreSQL-compatible relational database built for the cloud, offering high performance and availability.
NoSQL Databases:
DynamoDB: A fast and flexible NoSQL database service for all applications that need consistent, single-digit millisecond latency at any scale.
Analytical Databases:
Redshift: A fully managed, petabyte-scale data warehouse service for analytical workloads.
AWS Data Pipeline: Helps move data between different AWS data stores and on-premises data sources.
AWS Quicksite: A business intelligence service for creating visualizations and dashboards.

4. Front-end and Storage Services
S3 (Simple Storage Service): An object storage service for storing and retrieving any amount of data from anywhere on the web. It can also host static websites.
Elastic Block Store (EBS): Provides block-level storage volumes for use with EC2 instances.
Elastic File Store (EFS): Provides scalable file storage for EC2 instances that can be shared across multiple instances
Route 53: A highly available and scalable cloud Domain Name System (DNS) web service.
CloudFront: A fast content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency.

5. Security and Networking
VPC (Virtual Private Cloud): Allows you to provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define.
Subnets: Allow you to divide your VPC into smaller, more manageable network segments. Public subnets allow direct internet access, while private subnets do not.
VPN (Virtual Private Network): Establishes an encrypted connection over the internet between your on-premises network (Data Centres) and your VPC.
Direct Connect: Provides a dedicated network connection from your premises to AWS.

6. Other Important Services and DevOps

SQS (Simple Queue Service): A fully managed message queuing service for asynchronous communication between application components.
SNS (Simple Notification Service): A highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and serverless applications.
DevOps:
AWS Pipeline: Helps automate continuous integration and continuous delivery (CI/CD) workflows.
CodeBuild: Compiles source code, runs tests, and produces software packages that are ready to deploy.
CodeDeploy: Automates code deployments to any instance, including EC2 instances and on-premises servers.
CloudFormation: An infrastructure as code (IaC) service that helps you model and set up your AWS resources.

EXAM QUESTIONS FOR AWS CLOUD PRACTITIONER (CLF-C02)

1. Which of the following AWS Global Infrastructure components is made up of one or more discrete data centers, each with redundant power, networking, and connectivity and housed in separate facilities?
A. Edge location
B. AWS Region
C. VPC
D. Availability Zone
Correct Answer: Option D - Availability Zone

Correct Option:
D. Availability Zone: An Availability Zone (AZ) is a component of AWS infrastructure that consists of one or more discrete data centers, each with independent power, networking, and connectivity, and housed in separate physical facilities. These are designed to be isolated from failures in other zones, while still being able to communicate with other AZs in the same region over low-latency networks. AZs provide high availability and fault tolerance.

Incorrect Options:
A. Edge location: Edge locations are not data centers for hosting your applications; instead, they are locations where AWS caches content (like from CloudFront) for faster delivery to users. They are typically used for improving latency for end users but do not have the same infrastructure setup as Availability Zones.
B. AWS Region: An AWS Region is a broader geographical area that consists of multiple Availability Zones. It does not refer to individual data centers but rather to a collection of AZs, ensuring fault isolation and better disaster recovery. Regions consist of multiple AZs for redundancy but are not individual data centers themselves.
C. VPC (Virtual Private Cloud): A VPC is a virtual network within AWS where users can launch AWS resources, such as EC2 instances. It is a logical construct for isolating and managing network resources but does not refer to physical data centers or redundancy mechanisms like those in Availability Zones.

2. Which Amazon EC2 instance purchasing option lets you take advantage of unused EC2 capacity in the AWS Cloud and provides up to a 90% discount compared to On-Demand prices?
A. Dedicated Host
B. Standard Reserved Instance
C. Spot Instance
D. Convertible Reserved Instance
Correct Answer: Option C - Spot Instance

Correct Option:
C: Spot Instances allow you to bid for unused EC2 capacity at a significantly lower price compared to On- Demand Instances, with discounts of up to 90%. These are ideal for flexible workloads, where your application can tolerate interruptions, as AWS may reclaim the instance when the capacity is no longer available or the Spot price exceeds your bid price.

Incorrect Options:
A. Dedicated Host: A Dedicated Host gives you a physical EC2 server dedicated for your use. It provides you with visibility and control over the physical server but does not offer any discounts comparable to Spot Instances. This option is used when you need compliance, regulatory requirements, or licensing flexibility, but it's more expensive than On-Demand pricing.

B. Standard Reserved Instance: Standard Reserved Instances offer discounts on EC2 instances in exchange for committing to use them for a 1- or 3-year term. While this offers significant savings (up to 72% compared to On-Demand), it's less flexible than Spot Instances and does not provide the steep discounts (up to 90%) that Spot Instances can offer.
D. Convertible Reserved Instance: Convertible Reserved Instances provide flexibility to change the instance type, operating system, or tenancy during the reservation term while still offering a discount compared to On-Demand pricing. The savings are lower than Spot Instances and typically up to 66%, not the 90% discount seen with Spot Instances.

3. A startup is developing a mobile app with a database service to store user data. The app is expected to grow rapidly, and the company needs a flexible and scalable database service that can handle unpredictable traffic and workload spikes. Which service should the startup use if they need a scalable, fast, and flexible non-relational database service?
A. Amazon RDS
B. Amazon Redshift
C. Amazon S3
D. Amazon DynamoDB
Correct Answer: Option D - Amazon DynamoDB

Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability. DynamoDB lets you offload the administrative burdens of operating and scaling a distributed database so that you don't have to worry about hardware provisioning, setup, and configuration, replication, software patching, or cluster scaling. DynamoDB also offers encryption at rest, which eliminates the operational burden and complexity involved in protecting sensitive data.

4. For security audit purposes, a company needs to download compliance-related documents in AWS such as ISO certifications, Payment Card Industry (PCI), and Service Organization Control (SOC) reports. Where can they retrieve these files?
A. WS Artifact
B. AWS Certificate Manager
C. AWS Trusted Advisor
D. AWS CloudTrail
Correct answer: Option A

AWS Artifact is your go-to, central resource for compliance-related information that matters to you. It provides on-demand access to AWS' security and compliance reports and select online agreements. Reports available in AWS Artifact include our Service Organization Control (SOC) reports, Payment Card Industry (PCI) reports, and certifications from accreditation bodies across geographies and compliance verticals that validate the implementation and operating effectiveness of AWS security controls.

5. Which statement below is correct regarding the components of the AWS Global Infrastructure?
A. An edge location contains multiple AWS Regions.
B. An Availability Zone contains edge locations.
C. An AWS Region contains multiple Availability Zones.
D. An Availability Zone contains multiple AWS Regions.
Correct Answer: Option C

The AWS Global Infrastructure delivers a cloud infrastructure companies can depend on—no matter their size, changing needs, or challenges. The AWS Global Infrastructure is designed and built to deliver the most flexible, reliable, scalable, and secure cloud computing environment with the highest quality global network performance available today. An Availability Zone gives customers the ability to operate production applications and databases that are more highly available, fault-tolerant, and scalable than would be possible from a single data centre. All AZs are interconnected with high-bandwidth, low-latency networking over fully redundant, dedicated metro fibre providing high-throughput, low-latency networking between AZs.

6. The engineering team at a leading e-commerce company is anticipating a surge in traffic because of a flash sale planned for the weekend. You have estimated the web traffic to be 10x. The content of your website is highly dynamic and changes very often. As a Solutions Architect, which of the following options would you recommend to make sure your infrastructure scales for that day?
A. Use a Route53 Multi Value record

B. Use a CloudFront distribution in front of your website
C. Use an Auto Scaling Group
D. Deploy the website on S3
Correct Answer: C

Use an Auto Scaling Group – An Auto Scaling group (ASG) contains a collection of Amazon EC2 instances that are treated as a logical grouping for automatic scaling and management. An Auto Scaling group also enables you to use Amazon EC2 Auto Scaling features such as health check replacements and scaling policies. Both maintaining the number of instances in an Auto Scaling group and automatic scaling are the core functionality of the Amazon EC2 Auto Scaling service. The size of an Auto Scaling group depends on the number of instances that you set as the desired capacity. You can adjust its size to meet demand, either manually or by using automatic scaling.

7. A leading e-commerce company runs its IT infrastructure on AWS Cloud. The company has a batch job running at 7 am daily on an RDS database. It processes shipping orders for the past day and usually gets around 2000 records that need to be processed sequentially in a batch job via a shell script. The processing of each record takes about 3 seconds. What platform do you recommend to run this batch job?
A. AWS Lambda
B. Amazon Kinesis Data Streams
C. Amazon EC2
D. AWS Glue
Correct Answer: C

Correct Option:
Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale Cloud computing easier for developers. Amazon EC2's simple web service interface allows you to obtain and configure capacity with minimal friction. It provides you with complete control of your computing resources and lets you run on Amazon's proven computing environment. AWS Batch can be used to plan, schedule, and execute your batch computing workloads on Amazon EC2 Instances.

Incorrect options:
- AWS Glue – AWS Glue is a fully managed extract, transform, and load (ETL) service that makes it easy for customers to prepare and load their data for analytics.
- AWS Glue job is meant to be used for batch ETL data processing. Glue is for performing ETL, but cannot run custom shell scripts and hence not the right choice here.
- Amazon Kinesis Data Streams – Amazon Kinesis Data Streams (KDS) is a massively scalable and durable real-time data streaming service. KDS can continuously capture gigabytes of data per second from hundreds of thousands of data sources.

8. The company has grown from a small startup to an enterprise employing over 1000 people. As the team size has grown, the company has recently observed some strange behavior, with S3 bucket settings being changed regularly.
How can you figure out what's happening without restricting the rights of the users?
A. Implement an IAM policy to forbid users to change S3 bucket settings
B. Implement a bucket policy requiring MFA for all operations
C. Use S3 access logs to analyze user access using Athena
D. Use CloudTrail to analyze API calls
Correct Answer: D

Correct Option:
Use CloudTrail to analyze API calls – AWS CloudTrail is a service that enables governance, compliance, operational auditing, and risk auditing of your AWS account. With CloudTrail, you can log, continuously monitor and retain account activity related to actions across your AWS infrastructure. CloudTrail provides the event history of your AWS account activity, including actions taken through the AWS Management Console, AWS SDKs, command-line tools, and other AWS services.
In general, to analyze any API calls made within an AWS account, CloudTrail is used.

Incorrect Options:
- Implement an IAM policy to forbid users to change S3 bucket settings – You manage access in AWS by creating policies and attaching them to IAM identities (users, groups of users, or roles) or AWS resources. A policy is an object in AWS that, when associated with an identity or resource, defines their permissions. AWS evaluates these policies when an IAM principal (user or role) makes a request.

- Permissions in the policies determine whether the request is allowed or denied. Most policies are stored in AWS as JSON documents. AWS supports six types of policies: identity-based policies, resource-based policies, permissions boundaries, AWS Organizations SCPs, ACLs, and session policies. Implementing an IAM policy to forbid users would be disruptive and wouldn't go unnoticed.
- Use S3 access logs to analyze user access using **Amazon S3 Server Access Logging** provides **detailed records of requests** made to objects within an S3 bucket. These logs can be analyzed using **Amazon Athena**, a serverless query service from Amazon Web Services.

9. A developer in your company has set up a classic 2-tier architecture consisting of an Application Load Balancer and an Auto Scaling group (ASG) managing a fleet of EC2 instances. The ALB is deployed in a subnet of size 10.0.1.0/24 and the ASG is deployed in a subnet of size 10.0.4.0/22.
As a solutions architect, you would like to adhere to the security pillar of the well-architected framework. How do you configure the security group of the EC2 instances to only allow traffic coming from the ALB?
A. Add a rule to authorize the security group of the ALB
B. Add a rule to authorize the CIDR 10.0.1.0/24
C. Add a rule to authorize the security group of the ASG
D. Add a rule to authorize the CIDR 10.0.4.0/22
Correct Answer: A

Correct Option:
An Auto Scaling group (ASG) contains a collection of Amazon EC2 instances that are treated as a logical grouping for automatic scaling and management. An Auto Scaling group also enables you to use Amazon EC2 Auto Scaling features such as health check replacements and scaling policies. Both maintaining the number of instances in an Auto Scaling group and automatic scaling are the core functionality of the Amazon EC2 Auto Scaling service.
Add a rule to authorize the security group of the ALB: A security group acts as a virtual firewall that controls the traffic for one or more instances. When you launch an instance, you can specify one or more security groups; otherwise, we use the default security group.

Incorrect Option:
Add a rule to authorize the CIDR 10.0.4.0/22
Add a rule to authorize the security group of the ASG
Add a rule to authorize the CIDR 10.0.1.0/24
Adding the entire CIDR of the ALB would work, but wouldn't guarantee that only the ALB can access the EC2 instances that are part of the ASG. Here, the right solution is to add a rule on the ASG security group to allow incoming traffic only from the security group configured for the ALB.

10. A company has migrated its application from a monolith architecture to a microservices-based architecture. The development team has updated the Route 53 simple record to point "myapp.mydomain.com" from the old Load Balancer to the new one. The users are still not redirected to the new Load Balancer. What has gone wrong in the configuration?
A. The TTL is still in effect
B. The Alias Record is misconfigured
C. The CNAME Record is misconfigured
D. The health checks are failing
Correct Answer: B

Q11) A company is using Amazon S3 to store their static media contents such as photos and videos. Which of the following should you use to provide specific users access to the bucket?
A. SSH key
B. Security Group
C. **Bucket Policy**
D. Network Access Control List
Correct Answer: Option C

Correct Option:
Bucket Policies in Amazon S3 allow you to define specific access permissions to your S3 buckets and objects. You can use these policies to grant or deny access to users based on their AWS account, IP address, or other conditions. This makes it the most suitable option for controlling access to the contents of an S3 bucket.

Incorrect Option:

- SSH keys are used for securely accessing instances (e.g., EC2 instances) and do not apply to granting or managing access to S3 buckets.
- Security Groups control the inbound and outbound traffic to instances and do not apply to Amazon S3, which is a managed service. They are used at the instance or service level, not for S3 bucket access.
- NACLs are used to control traffic at the subnet level within a VPC and do not apply to S3 bucket access, which is managed through IAM policies and bucket policies.

Q12) Which of the following services should you use to deploy and easily rollback a web application from your Git repository to your on-premises server?

A. AWS Systems Manager
B. AWS CloudFormation
C. AWS OpsWorks (Correct)
D. AWS Elastic Beanstalk
Correct Answer: Option C

Correct Option:
AWS OpsWorks allows you to deploy applications and manage servers, both on-premises and on AWS. It integrates with Git repositories, enabling you to deploy changes to your applications easily. Additionally, OpsWorks provides a simple rollback mechanism, allowing you to revert to previous versions of your application if needed.

Incorrect Option:

- AWS Systems Manager: It is primarily used for managing and automating operational tasks on AWS resources and on-premises servers. It doesn't directly facilitate application deployment and rollback from a Git repository.
- AWS CloudFormation: It is used to define and provision AWS infrastructure using code. While it helps automate resource deployment, it is not intended for deploying application code directly from a Git repository to servers.
- AWS Elastic Beanstalk is a **Platform as a Service (PaaS)** offering from Amazon Web Services that simplifies application deployment and management in the cloud.

EXAM QUESTIONS FOR AWS SOLUTION ARCHITECT ASSOCIATE (SAA-C03)

1. An online hypermarket company has deployed a web application using REST API with Amazon API Gateway. Recently they have upgraded the backend to make API scalable. After the upgrade, it was found that some of the consumers using older methods cannot access this API. The older method used by these consumers is not compatible with the responses by the backend host. How should a solution architect redesign the API Gateway to make API compatible with the old method?

A. Enable API caching in Amazon API Gateway
B. Configure Mapping templates with Amazon API Gateway
C. Set up Gateway Response customization in OpenAPI
D. Set up a method response model with Amazon API Gateway
Correct Answer: Option B -Configure Mapping templates with Amazon API Gateway
Mapping templates in Amazon API Gateway allow you to transform the request and response payloads. By using mapping templates, you can modify the API's response to make it compatible with older client methods, ensuring backward compatibility. This allows you to change the response format to match what the older clients expect, even if the backend service has been updated.

Incorrect Options:

- Enable API caching in Amazon API Gateway: API caching is used to improve the performance of your API by storing the responses for a period of time. It does not help in transforming or modifying the responses to make them compatible with older clients.
- Set up Gateway Response customization in OpenAPI: Gateway response customization allows you to modify error responses, such as 4XX and 5XX errors, returned by API Gateway. It does not handle transforming the response body to ensure compatibility with older clients.
- Set up a method response model with Amazon API Gateway: Method response models are used to define the data structure of the responses. They help validate the response payload but do not modify or transform the actual data sent back to the client. They cannot be used to make the API response compatible with older client methods.

2. An IT company uses Scaling policies to maintain an exact number of Amazon EC2 instances in different Availability zones as per application workloads. The developer team has developed a new version of the application for which a new

AMI needs to be updated to all the instances. For this, you have been asked to ensure that instances with previous AMI are phased out quickly. Which termination criteria best suits the requirement?
A. Specify termination criteria using “ClosestToNextInstanceHour” predefined termination policy
B. Specify termination criteria using “OldestInstance” predefined termination policy
C. Specify termination criteria using “AllocationStrategy” predefined termination policy
D. Specify termination criteria using “OldestLaunchTemplate” predefined termination policy
Correct Answer: Option D

Specify termination criteria using “OldestLaunchTemplate” predefined termination policy -
The "OldestLaunchTemplate" termination policy is used to terminate instances launched with the oldest launch template or configuration first. This ensures that instances running the previous AMI are terminated quickly, allowing the Auto Scaling group to replace them with instances using the new AMI.

3. A start-up company uses Amazon CloudFormation templates to launch Amazon EC2 instances in different regions. AMI used for these instances differs for each instance type and region. Separate templates need to be created to specify AMI ID as per instance type and as per the region in which instances need to be launched. The IT team is looking for an effective solution to updating the CloudFormation template with the correct AMI ID reducing additional management work. Which solution can be suggested to get the AMI ID updated in the most effective manner?
A. Use Custom resources and a Lambda function to create a function that will update AMI IDs in the CloudFormation template
B. Map AMI IDs to the specific instance type and regions. Manually update the AMI IDs in the CloudFormation templates
C. Use Custom resources along with Amazon SNS which will update AMI IDs in the CloudFormation template
D. Use Custom resources along with Amazon SQS which will update AMI IDs in the CloudFormation template
Correct Answer: Option A
Use Custom resources and a Lambda function to create a function that will update AMI IDs in the CloudFormation template-
Custom resources in CloudFormation allow you to extend the functionality of templates by invoking AWS Lambda functions. By using a Lambda function, you can dynamically fetch the latest AMI ID for a specific region and instance type at the time the CloudFormation stack is created or updated. This eliminates the need for manually updating the AMI IDs in the template and ensures that the instances always use the correct and up-to-date AMI.

Incorrect Options:
- Map AMI IDs to the specific instance type and regions. Manually update the AMI IDs in the CloudFormation templates:

Manually updating AMI IDs in CloudFormation templates increases the management overhead. It is error-prone and not scalable, especially when dealing with multiple regions and instance types.

- Use Custom resources along with Amazon SNS which will update AMI IDs in the CloudFormation template:

Amazon SNS (Simple Notification Service) is used for messaging and does not have the capability to update CloudFormation templates directly. While it can trigger actions, it is not intended for retrieving and updating AMI IDs dynamically in CloudFormation.

- Use Custom resources along with Amazon SQS which will update AMI IDs in the CloudFormation template:

Amazon SQS (Simple Queue Service) is a message queuing service and cannot be used to directly update CloudFormation templates. It is used for decoupling and buffering messages, not for handling dynamic AMI ID retrieval or updates in CloudFormation.

4. A start-up firm has established hybrid connectivity from an on-premises location to the AWS cloud using AWS Site-to-Site VPN. A large number of applications are deployed in the AWS cloud to be accessed from the on- premises location. Users are complaining of the slowness while accessing these applications during peak hours. You have been assigned to work on a solution to improve connectivity throughput from on-premises to AWS.
What solution can be designed to increase VPN throughput?
A. Establish multiple VPN connections to the ECMP-enabled Transit gateway. Enable dynamic routing on the Transit Gateway
B. Establish multiple VPN connections to ECMP-enabled Virtual Private gateway. Enable route propagation on the Virtual Private Gateway
C. Establish multiple VPN connections to multiple Transit gateways. Enable dynamic routing on the Transit Gateway
D. Establish multiple VPN connections to multiple Virtual private gateways. Enable route propagation on the Virtual Private Gateway
Correct Answer: Option B

Establish multiple VPN connections to an ECMP-enabled Virtual Private Gateway. Enable route propagation on the Virtual Private Gateway -

By establishing multiple VPN connections to a Virtual Private Gateway (VGW) and enabling Equal Cost Multi-Path (ECMP) routing, you can distribute traffic across multiple VPN tunnels, increasing overall throughput. Route propagation with dynamic routing allows for more efficient traffic management and failover, enhancing the performance and reliability of the connection.

Incorrect Options:

- Establish multiple VPN connections to the ECMP-enabled Transit Gateway. Enable dynamic routing on the Transit Gateway

While Transit Gateway supports ECMP, this option is less efficient compared to using a Virtual Private Gateway when it comes to simple site-to-site VPN connections. Transit Gateway is generally used for more complex network architectures involving multiple VPCs and on-premises networks.

- Establish multiple VPN connections to multiple Transit Gateways. Enable dynamic routing on the Transit Gateway

Using multiple Transit Gateways for a single on-premises connection adds unnecessary complexity and

5. An oil drilling company is planning to use Kubernetes clusters on offshore platforms. Some of these platforms are in remote locations without having any Internet access. The IT Team is looking for an automated option for cluster management along with the creation of clusters at the offshore platforms. What design can be proposed to manage these Kubernetes clusters?

A. Deploy Amazon EKS Anywhere using VMware vSphere. Use the EKS distro along with open-source tools for running the clusters

B. Deploy Amazon EKS Anywhere on AWS Outposts. Use Amazon EKS for running clusters

C. Deploy Amazon EKS using BareMetal deployments. Use Amazon EKS for running clusters

D. Deploy Amazon EKS Anywhere using AWS ECR. Use the EKS distro along with open-source tools for running the clusters

Correct Answer: Option A

Deploy Amazon EKS Anywhere using VMware vSphere. Use the EKS distro along with open-source tools for running the clusters - Amazon EKS Anywhere allows you to create and operate Kubernetes clusters on-premises without requiring an internet connection. When deployed on VMware vSphere, it provides a self-managed Kubernetes environment that can be operated offline, using the EKS Distro (the same Kubernetes distribution used by Amazon EKS). This is suitable for remote and disconnected environments like offshore platforms, where internet access may be limited or unavailable.

Incorrect Options:

- Deploy Amazon EKS Anywhere on AWS Outposts. Use Amazon EKS for running clusters - AWS Outposts provides on-premises infrastructure for AWS services, but it still requires an internet connection to the AWS Region for management and updates. This makes it unsuitable for completely disconnected environments like offshore platforms without internet access.

6. As a solutions architect, you have created a solution that utilizes an Application Load Balancer with stickiness and an Auto Scaling Group (ASG). The ASG spawns across 2 Availability Zones (AZ). AZ-A has 3 EC2 instances and AZ-B has 4 EC2 instances. The ASG is about to go into a scale-in event due to the triggering of a CloudWatch alarm. What will happen under the default ASG configuration?

A. An instance in the AZ-A will be created

B. The instance with the oldest launch configuration will be terminated in AZ-B

C. A random instance will be terminated in AZ-B

D. A random instance in the AZ-A will be terminated

Correct Answer: B

The instance with the oldest launch configuration will be terminated in AZ-B – Amazon EC2 Auto Scaling helps you ensure that you have the correct number of Amazon EC2 instances available to handle the load for your application. You create collections of EC2 instances, called Auto Scaling groups. You can specify the minimum number of instances in each Auto Scaling group, and Amazon EC2 Auto Scaling ensures that your group never goes below this size.

With each Auto Scaling group, you can control when it adds instances (referred to as scaling out) or removes instances (referred to as scaling in) from your network architecture. The default termination policy is designed to help ensure that your instances span Availability Zones evenly for high availability. The default policy is kept generic and flexible to cover a range of scenarios.

The default termination policy behavior is as follows:

1. Determine which Availability Zones have the most instances and at least one instance that is not protected from scale-in.
2. Determine which instances to terminate to align the remaining instances to the allocation strategy for the On-Demand or Spot Instance that is terminating.
3. Determine whether any of the instances use the oldest launch template or configuration:

3.a. Determine whether any of the instances use the oldest launch template unless there are instances that use a launch configuration.

3.b. Determine whether any of the instances use the oldest launch configuration.

4. After applying all of the above criteria, if there are multiple unprotected instances to terminate, determine which instances are closest to the next billing hour.
Per the given use-case, AZs will be balanced first, then the instance with the oldest launch configuration within that AZ will be terminated.
Incorrect Options:

- A random instance in the AZ-A will be terminated
- An instance in the AZ-A will be created
- A random instance will be terminated in AZ-B

These three options contradict the details provided in the explanation above. Hence these are incorrect.

7. A Big Data analytics company wants to set up an AWS cloud architecture that throttles requests in case of sudden traffic spikes. The company is looking for AWS services that can be used for buffering or throttling to handle such traffic variations.Which of the following services can be used to support this requirement?
A. Amazon API Gateway, Amazon SQS and Amazon Kinesis
B. Amazon Gateway Endpoints, Amazon SQS and Amazon Kinesis
C. Amazon SQS, Amazon SNS and AWS Lambda
D. Elastic Load Balancer, Amazon SQS, AWS Lambda
Correct option: Amazon API Gateway, Amazon SQS and Amazon Kinesis

8. Your company runs a web portal to match developers to clients who need their help. As a solutions architect, you've designed the architecture of the website to be fully serverless with API Gateway & AWS Lambda. The backend uses a DynamoDB table. You would like to automatically congratulate your developers on important milestones, such as – their first paid contract. All the contracts are stored in DynamoDB. Which DynamoDB feature can you use to implement this functionality such that there is the LEAST delay in sending automatic notifications?
A. DynamoDB Streams + Lambda
B. DynamoDB DAX + API Gateway
C. CloudWatch Events + Lambda
D. Amazon SQS + Lambda
Correct Answer: A

Amazon DynamoDB is a key-value and document database that delivers single-digit millisecond performance at any scale. It's a fully managed, multi-region, multi-master, durable database with built-in security, backup and restore, and in-memory caching for internet-scale applications.
DynamoDB Streams + Lambda – A DynamoDB stream is an ordered flow of information about changes to items in a DynamoDB table. When you enable a stream on a table, DynamoDB captures information about every modification to data items in the table. Whenever an application creates, updates, or deletes items in the table, DynamoDB Streams writes a stream record with the primary key attributes of the items that were modified.

9. A media company runs a photo-sharing web application that is accessed across three different countries. The application is deployed on several Amazon EC2 instances running behind an Application Load Balancer. With new government regulations, the company has been asked to block access from two countries and allow access only from the home country of the company. Which configuration should be used to meet this changed requirement?
A. Configure AWS WAF on the Application Load Balancer in a VPC
B. Configure the security group for the EC2 instances
C. Use Geo Restriction feature of Amazon CloudFront in a VPC
D. Configure the security group on the Application Load Balancer
Correct option: Configure AWS WAF on the Application Load Balancer in a VPC

AWS WAF is a web application firewall service that lets you monitor web requests and protect your web applications from malicious requests.
Use AWS WAF to block or allow requests based on conditions that you specify, such as the IP addresses.
You can also use AWS WAF preconfigured protections to block common attacks like SQL injection or cross-site scripting.
Configure AWS WAF on the Application Load Balancer in a VPC
You can use AWS WAF with your Application Load Balancer to allow or block requests based on the rules in a web access control list (web ACL). Geographic (Geo) Match Conditions in AWS WAF allows you to use AWS WAF to restrict application access based on the geographic location of your viewers.

10. A company has built a serverless application using API Gateway and AWS Lambda. The backend is leveraging an RDS Aurora MySQL database. The web application was initially launched in the Americas and the company would now like to expand it to Europe, where a read-only version will be available to improve latency. You plan on deploying the API

Gateway and AWS Lambda using CloudFormation but would like to have a read-only copy of your data in Europe as well. As a Solutions Architect, what do you recommend?
A. Use a DynamoDB Streams
B. Create a Lambda function to periodically back up and restore the Aurora database in another region
C. Use Aurora Read Replicas
D. Use Aurora Multi-AZ
Correct Options: C

Use Aurora Read Replicas - Amazon Aurora features a distributed, fault-tolerant, self-healing storage system that auto-scales up to 64TB per database instance. It delivers high performance and availability with up to 15 low-latency read replicas, point-in-time recovery, continuous backup to Amazon S3, and replication across three Availability Zones (AZs).
Aurora Replicas are independent endpoints in an Aurora DB cluster, best used for scaling read operations. Up to 15 Aurora Replicas can be distributed across the Availability Zones that a DB cluster spans within an AWS Region. The DB cluster volume is made up of multiple copies of the data for the DB cluster. However, the data in the cluster volume is represented as a single, logical volume to the primary instance and Aurora Replicas in the DB cluster. You can also set up two Aurora MySQL DB clusters in different AWS Regions, by creating an Aurora Read Replica of an Aurora MySQL DB cluster in a different AWS Region. In this way, Aurora Read Replicas can be deployed globally.

Incorrect Options:
- Use Aurora Multi-AZ – Aurora, stores copies of the data in a DB cluster across multiple Availability Zones in a single AWS Region, regardless of whether the instances in the DB cluster span multiple Availability Zones. When data is written to the primary DB instance, Aurora synchronously replicates the data across Availability Zones to six storage nodes associated with your cluster volume. Doing so provides data redundancy, eliminates I/O freezes, and minimizes latency spikes during system backups. Always remember that the main purpose for multi-AZ is high availability whereas the main purpose of read replicas is read scalability.
- Use DynamoDB Streams – DynamoDB Streams is a powerful service that you can combine with other AWS services to solve many problems. When enabled, DynamoDB Streams captures a time-ordered sequence of item-level modifications in a DynamoDB table and durably stores the information for up to 24 hours. Applications can access a series of stream records, which contain an item change, from a DynamoDB stream in near real-time.
- Create a Lambda function to periodically back up and restore the Aurora database in another region –Lambda can be used to create backups for RDS. However, the process is not an optimized solution, especially when Aurora already offers the read replica feature

11. A company has noticed that its EBS storage volume (io1) accounts for 90% of the cost and the remaining 10% cost can be attributed to the EC2 instance. The CloudWatch metrics report that both the EC2 instance and the EBS volume are under-utilized. The CloudWatch metrics also show that the EBS volume has occasional I/O bursts. The entire infrastructure is managed by AWS CloudFormation. As a Solutions Architect, what do you propose to reduce the costs?
A. Keep the EBS volume to io1 and reduce the IOPS
B. Convert the Amazon EC2 instance EBS volume to gp2
C. Don't use a CloudFormation template to create the database as the CloudFormation service incurs greater service charges
D. Change the Amazon EC2 instance type to something much smaller
Correct Options: B

Amazon EBS provides various volume types that differ in performance characteristics and price so that you can tailor your storage performance and cost to the needs of your applications. The volume types fall into two categories:
SSD-backed volumes optimized for transactional workloads involving frequent read/write operations with small I/O size, where the dominant performance attribute is IOPS.
HDD-backed volumes optimized for large streaming workloads where throughput (measured in MiB/s) is a better performance measure than IOPS.
Provisioned IOPS SSD (io1) volumes are designed to meet the needs of I/O-intensive workloads, particularly database workloads, that are sensitive to storage performance and consistency. Unlike gp2, which uses a bucket and credit model to calculate performance, an io1 volume allows you to specify a consistent IOPS rate when you create the volume, and Amazon EBS delivers the provisioned performance 99.9 percent of the time.
Convert the Amazon EC2 instance EBS volume to gp2 – General Purpose SSD (gp2) volumes offer cost-effective storage that is ideal for a broad range of workloads.
These volumes deliver single-digit millisecond latencies and the ability to burst to 3,000 IOPS for an extended duration. Between a minimum of 100 IOPS (at 33.33 GiB and below) and a maximum of 16,000 IOPS (at 5,334 GiB and above), baseline performance scales linearly at 3 IOPS per GiB of volume size. AWS designs gp2 volumes to deliver a provisioned performance of 99% uptime. A gp2 volume can range in size from 1 GiB to 16 TiB.
Therefore, gp2 is the right choice as it is more cost-effective than io1, and it also allows a burst in performance when needed.

Incorrect Options:

- Keep the EBS volume to io1 and reduce the IOPS – Keeping the EBS volume to io1 and reducing the IOPS may interfere with the burst of performance we need, so this option is ruled out.
- Change the Amazon EC2 instance type to something much smaller – Changing the EC2 instance type to something much smaller won't affect 90% of the costs that are incurred, therefore this option is also incorrect.
- Don't use a CloudFormation template to create the database as the CloudFormation service incurs greater service charges – This statement is incorrect as CloudFormation is a free service to use. The resources that are invoked by CloudFormation are charged as per their utilization rates, but using CloudFormation will not cost anything.

12. An e-commerce company wants to migrate its on-premises application to AWS. The application consists of application servers and a Microsoft SQL Server database. The solution should result in the maximum possible availability for the database layer while minimizing operational and management overhead.
As a solutions architect, which of the following would you recommend to meet the given requirements?
A. Migrate the data to Amazon RDS for SQL Server database in a cross-region Multi-AZ deployment
B. Migrate the data to Amazon RDS for SQL Server database in a cross-region read-replica configuration
C. Migrate the data to an EC2 instance hosted SQL Server database. Deploy the EC2 instances in a Multi-AZ configuration
D. Migrate the data to Amazon RDS for SQL Server database in a Multi-AZ deployment
Correct Options: D

Migrate the data to Amazon RDS for SQL Server database in a Multi-AZ deployment -
Amazon RDS supports Multi-AZ deployments for Microsoft SQL Server by using either SQL Server Database Mirroring (DBM) or Always On Availability Groups (AGs). Amazon RDS monitors and maintains the health of your Multi-AZ deployment. If problems occur, RDS automatically repairs unhealthy DB instances, reestablishes synchronization, and initiates failovers.
Multi-AZ deployments provide increased availability, data durability, and fault tolerance for DB instances. In the event of planned database maintenance or unplanned service disruption, Amazon RDS automatically fails over to the up-to-date secondary DB instance. This functionality lets database operations resume quickly without manual intervention. The primary and standby instances use the same endpoint, whose physical network address transitions to the secondary replica as part of the failover process. You don't have to reconfigure your application when a failover occurs.
This option provides the maximum possible availability for the database layer while minimizing operational and management overhead.

Incorrect Options:

- Migrate the data to an EC2 instance hosted SQL Server database. Deploy the EC2 instances in a Multi-AZ configuration – Hosting SQL Server database on EC2 instance involves significant operational and management overhead in terms of OS patching, database patching, etc. So this option is incorrect.
- Migrate the data to Amazon RDS for SQL Server database in a cross-region read-replica configuration – Amazon RDS Read Replicas enables you to create one or more read-only copies of your database instance within the same AWS Region or in a different AWS Region. Read replicas are used to enhance the read scalability of a database. You cannot use read replicas to improve the availability of a database. Therefore this option is incorrect.
- Migrate the data to Amazon RDS for SQL Server database in a cross-region Multi-AZ deployment – Amazon RDS Multi-AZ deployments provide enhanced availability for database instances within a single AWS Region. There is no such thing as a cross-region Multi-AZ deployment. Hence this option is incorrect.

EXAM QUESTIONS FOR AWS DEVOPS PROFESSIONAL (DOP-C02)

Q1) You are a Devops Engineer for your company. The company has a number of Cloudformation templates in AWS. There is a concern from the IT Security department and they want to know who all use the Cloudformation stacks in the company's AWS account. Which of the following can be done to take care of this security concern?
A. Enable Cloudwatch events for each cloudformation stack to track the resource creation events
B. Enable Cloudtrail logs so that the API calls can be recorded
C. Enable Cloudwatch logs for each cloudformation stack to track the resource creation events
D. Connect SQS and Cloudformation so that a message is published for each resource created in the Cloudformation stack
Correct Answer: Option B
Reason: AWS CloudTrail records API calls made on your account, including those made by CloudFormation. It provides detailed logs of who accessed or modified CloudFormation stacks, offering visibility into actions taken by users and services. This information helps address security concerns by allowing tracking of who uses and modifies CloudFormation stacks.

Incorrect Options:

- Enable CloudWatch events for each CloudFormation stack to track the resource creation events

CloudWatch Events can monitor resource creation events, but it doesn't provide detailed information on who initiated the CloudFormation stack or the API calls made. It focuses more on resource state changes rather than user activity tracking.

- Enable CloudWatch logs for each CloudFormation stack to track the resource creation events

CloudWatch Logs can track logs from various AWS services, but they don't specifically track CloudFormation API calls or user actions related to stack usage. They are more suited for application and resource monitoring rather than tracking user activity in CloudFormation.

- Connect SQS and CloudFormation so that a message is published for each resource created in theCloudFormation stack

Amazon SQS can be used to send messages when resources are created, but it does not track the user or the API calls made. It is designed for decoupling components in a distributed system rather than monitoring and logging API activity.

Q2) Your development team uses .Net for their web application. The application code is hosted in a Git repository. They want to set up continuous integration and deployment pipelines and deploy it to AWS. Which of the following methods can be used to fulfil this requirement?

A. Use the Elastic Beanstalk service to provision an IIS platform web environment to host the application in AWS
B. Use the Code Pipeline service to set up the pipelines for the application in AWS
C. Create a source bundle for the .Net code and upload it as an application revision
D. Use a chef recipe to deploy the code and attach it to an Elastic Beanstalk environment

Correct Answer: B

Reason: AWS CodePipeline is a fully managed continuous integration and continuous delivery (CI/CD) service that automates the building, testing, and deployment of code. It integrates seamlessly with Git repositories and can work with .NET applications, making it the most suitable option for setting up CI/CD pipelines for a .NET web application hosted in a Git repository.

Incorrect Options:

- Use the Elastic Beanstalk service to provision an IIS platform web environment to host the application in AWS

While Elastic Beanstalk can host .NET applications on an IIS platform, it is not a CI/CD tool. Elastic Beanstalk provides an environment for deploying and running applications, but it does not inherently provide pipeline creation or management for continuous integration and deployment.

- Create a source bundle for the .NET code and upload it as an application revision

Creating a source bundle and uploading it as an application revision can be part of the deployment process but does not establish an automated CI/CD pipeline. It involves manual steps and lacks the automation that CodePipeline offers.

- Use a Chef recipe to deploy the code and attach it to an Elastic Beanstalk environment

Chef recipes can be used for OS-level configuration, Installing packages, Managing system settings

Q3) Your company is planning on using the available services in AWS to completely automate their integration, build and deployment process. They are planning on using AWS CodeBuild to build their artifacts. When using CodeBuild, which of the following files specifies a collection of build commands that can be used by the service during the build process.

A. appspec.yml
B. buildspec.yml
C. buildspec.xml
D. appspec.json

Correct Answer: Option B

Reason: buildspec.yml is the default file that AWS CodeBuild looks for during the build process. It contains a collection of build commands and related settings in YAML format that CodeBuild uses to run the build process.

Incorrect Options:

- Appspec.yml - appspec.yml is used by AWS CodeDeploy, not CodeBuild. It specifies how CodeDeploy manages deployment actions for an application, not how to build the code.
- Buildspec.xml- CodeBuild does not support XML format for build specification files. CodeBuild expects the build commands and phases to be defined in a YAML file,specifically named buildspec.yml.

Q4) You are a Devops engineer for your company. The company hosts a web application that is hosted on a single EC2 Instance. The end users are complaining of slow response times for the application. Which of the following can be used to effectively scale the application?
A. Use Auto Scaling Groups to launch multiple instances and place them behind an ELB
B. Use Auto Scaling launch configurations to launch multiple instances and place them behind an ELB
C. Use Amazon RDS with the Multi-AZ feature
D. Use Cloudformation to deploy the app again with an Amazon RDS with the Multi-AZ feature
Correct Answer: Option A
Reason: Auto Scaling Groups (ASGs) allow you to automatically launch and manage multiple EC2 instances based on the demand for your application. By placing these instances behind an Elastic Load Balancer (ELB), you distribute incoming traffic evenly across all instances, ensuring higher availability and better response times for the application.

Incorrect Options:
- Use Auto Scaling launch configurations to launch multiple instances and place them behind an ELB

Launch configurations are used as templates for Auto Scaling Groups but do not themselves provide scaling capabilities. You still need an Auto Scaling Group to handle the scaling process and manage instances.
- Use Amazon RDS with the Multi-AZ feature

Amazon RDS Multi-AZ provides high availability for database instances but does not address application performance issues directly. It does not help in scaling the web application itself.
- Use CloudFormation to deploy the app again with an Amazon RDS with the Multi-AZ feature

CloudFormation can automate the deployment of resources, but adding Amazon RDS with Multi-AZ only improves database availability, not the scaling or performance of the web application running on the EC2 instance.

Q5) You are using Jenkins as your continuous integration systems for the application hosted in AWS. The builds are then placed on newly launched EC2 Instances. You want to ensure that the overall cost of the entire continuous integration and deployment pipeline is minimized. Which of the below options would meet these requirements? Choose 2 answers from the options given below:
A. Ensure that all build tests are conducted using Jenkins before deploying the build to newly launched EC2 Instances
B. Ensure that all build tests are conducted on the newly launched EC2 Instances
C. Ensure the Instances are launched only when the build tests are completed
D. Ensure the Instances are created beforehand for faster turnaround time for the application builds to be placed
Correct Answer: Option A and C
Reason:
- Ensure that all build tests are conducted using Jenkins before deploying the build to newly launched EC2 Instances

Running build tests directly in Jenkins ensures that only successful builds are deployed to EC2 instances. This minimizes the need for unnecessary EC2 instances, reducing costs since instances are only launched when needed.
- Ensure the Instances are launched only when the build tests are completed

Launching instances only after build tests are successful prevents the cost of running EC2 instances during the testing phase. This way, you only incur costs for the deployment phase, optimizing resource usage and cost.

Incorrect Options:
- Ensure that all build tests are conducted on the newly launched EC2 Instances

Running build tests on EC2 instances means the instances are active during the entire testing and deployment process, increasing costs since you pay for the running time of these instances even if the build fails.
- Ensure the Instances are created beforehand for faster turnaround time for the application builds to be placed

Pre-creating instances leads to higher costs because you pay for idle instances waiting to be used. This approach doesn't take advantage of auto-scaling and on-demand instance provisioning, which can minimize costs.

Q6) As an e-sport tournament hosting company, you have servers that need to scale and be highly available. Therefore you have deployed an Elastic Load Balancer (ELB) with an Auto Scaling group (ASG) across 3 Availability Zones (AZs). When e-sport tournaments are running, the servers need to scale quickly. And when tournaments are done, the servers can be idle. As a general rule, you would like to be highly available and can scale and optimize your costs. What do you recommend? (Select two)
A. Use Dedicated hosts for the minimum capacity

B. Set the minimum capacity to 3
C. Use Reserved Instances for the minimum capacity
D. Set the minimum capacity to 2
E. Set the minimum capacity to 1
Correct Answer: C, D
Explanation:
Set the minimum capacity to 2
An Auto Scaling group contains a collection of Amazon EC2 instances that are treated as a logical grouping for automatic scaling and management. An Auto Scaling group also enables you to use Amazon EC2 Auto Scaling features such as health check replacements and scaling policies. Both maintaining the number of instances in an Auto Scaling group and automatic scaling are the core functionality of the Amazon EC2 Auto Scaling service.
You configure the size of your Auto Scaling group by setting the minimum, maximum, and desired capacity.

Incorrect Options:

- Set the minimum capacity to 1 – This is not failure-proof, since only one instance will be maintained consistently and this will be from only one availability zone.
- Set the minimum capacity to 3 – This is not a cost-effective option, as two instances in two different AZs are enough to make the architecture disaster-proof.
- Use Dedicated hosts for the minimum capacity – As there is no use-case to utilize existing per-socket, per-core, or per-VM software licenses or to run the instance on a dedicated physical host, so the option to usededicated hosts for the minimum capacity is ruled out.

Q7) A company runs a popular dating website on the AWS Cloud. As a Solutions Architect, you've designed the architecture of the website to follow a serverless pattern on the AWS Cloud using API Gateway and AWS Lambda. The backend uses an RDS PostgreSQL database. Currently, the application uses a username and password combination to connect the Lambda function to the RDS database. You would like to improve the security at the authentication level by leveraging short-lived credentials. What will you choose? (Select two)
A. Embed a credential rotation logic in the AWS Lambda, retrieving them from SSM
B. Attach an AWS Identity and Access Management (IAM) role to AWS Lambda
C. Restrict the RDS database security group to the Lambda's security group
D. Use IAM authentication from Lambda to RDS PostgreSQL
E. Deploy AWS Lambda in a VPC
Correct Answer: B, D
Explanation:
You can authenticate to your DB instance using AWSIdentity and Access Management (IAM) database authentication. IAM database authentication works with MySQL and PostgreSQL. With this authentication method, you don't need to use a password when you connect to a DB instance. Instead, you use an authentication token.

Q9) For security purposes, a development team has decided to deploy the EC2 instances in a private subnet. The team plans to use VPC endpoints so that the instances can access some AWS services securely. The members of the team would like to know about the two AWS services that support Gateway Endpoints. As a solutions architect, which of the following services would you suggest for this requirement? (Select two)
A. Amazon Kinesis
B. Amazon Simple Notification Service (SNS)
C. Amazon Simple Queue Service (SQS)
D. Amazon S3
E. DynamoDB
Correct Answer: D, E

DynamoDB
A VPC endpoint enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by AWS PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. Instances in your VPC do not require public IP addresses to communicate with resources in the service. Traffic between your VPC and the other service does not leave the Amazon network. Endpoints are virtual devices. They are horizontally scaled, redundant, and highly available VPC

components. They allow communication between instances in your VPC and services without imposing availability risks or bandwidth constraints on your network traffic.

Incorrect Options:
A. Amazon Kinesis: Kinesis does not support Gateway Endpoints. To access Kinesis securely from a VPC, you would need to use an Interface VPC Endpoint (powered by AWS PrivateLink), not a Gateway Endpoint.
B. Amazon Simple Notification Service (SNS): SNS also does not support Gateway Endpoints. Like Kinesis, you would need an Interface VPC Endpoint to securely connect to SNS from a private subnet.
C. Amazon Simple Queue Service (SQS): SQS doesn't support Gateway Endpoints. Access to SQS from within a VPC requires an Interface VPC Endpoint, similar to SNS and Kinesis.

Q10) A government agency has a VMware-based automated server build system on their on-premises network which uses a virtualization software that allows them to create server images of their application. They instructed their DevOps Engineer to set up a system that will allow them to test their server images using their on-premises server pipeline to resemble the build and behavior on Amazon EC2. In this way, the agency can verify the functionality of their application, detect incompatibility issues, and determine any prerequisites on the new Amazon Linux 2 operating system that will be used in AWS. Which of the following solutions should the DevOps Engineer implement to accomplish this task?
A. Launch an Amazon EC2 instance with the latest Amazon Linux OS in AWS. Use the AWS VM Import/Export service to import the EC2 image, export it to a VMware ISO in an S3 bucket, and then import the ISO to an on-premises server. Once done, commence the testing activity to verify the application's functionalities.
B. Download the latest AmazonLinux2.iso of the Amazon Linux 2 operating system and import it to your on-premises network. Directly launch a new on-premises server based on the imported ISO, without any virtual platform. Deploy the application, and commence testing.
C. Launch a new on-premises server with any distribution of Linux operating system such as CentOS, Ubuntu or Fedora since these are technically the same. Deploy the application to the server for testing.
D. Use an AWS OpsWorks deployment agent that can reformat the target server on-premises to use Amazon Linux and then install the application afterward. Start the testing once the installation is complete.
Correct Answer: A

A. AWS VM Import/Export service:
This solution leverages the AWS VM Import/Export service to import an Amazon EC2 instance image based on the Amazon Linux 2 OS into a VMware-compatible format. This allows testing of the application on an on-premises server, mirroring the EC2 environment and ensuring compatibility with Amazon Linux 2. This provides a highly realistic environment for testing and validation.

Incorrect Options:
B. Download AmazonLinux2.iso:
While this option directly runs Amazon Linux 2 on an on-premises server, it lacks the integration with VMware's virtualization platform that the government agency's existing automated build system uses. This wouldn't replicate the build and behavior of EC2 as required.
C. Use other Linux distributions (CentOS, Ubuntu, Fedora):
Even though CentOS, Ubuntu, and Fedora are Linux- based operating systems, they are not the same as Amazon Linux 2, which has specific optimizations and configurations tailored for AWS. Using these distributions would not help detect Amazon Linux 2- specific compatibility issues.
D. AWS OpsWorks deployment agent:
AWS OpsWorks is a configuration management service and is not designed for transforming on-premises servers to Amazon Linux 2. It cannot reformat a target server into an Amazon Linux instance, so it wouldn't meet the requirement of closely simulating the EC2 environment on-premises.

Q11) A company has developed an AWS Lambda function that handles orders received through an API. The company is using AWS CodeDeploy to deploy the Lambda function as the final stage of a CI/CD pipeline.
A DevOps Engineer has noticed there are intermittent failures of the ordering API for a few seconds after deployment. After some investigation, the DevOps Engineer believes the failures are due to database changes not having fully propagated before the Lambda function begins executing. How should the DevOps Engineer overcome this?
A. Add a BeforeAllowTraffic hook to the AppSpec file that tests and waits for any necessary database changes before traffic can flow to the new version of the Lambda function
B. Add an AfterAllowTraffic hook to the AppSpec file that forces traffic to wait for any pending database changes before allowing the new version of the Lambda function to respond

C. Add a BeforeInstall hook to the AppSpec file that tests and waits for any necessary database changes before deploying the new version of the Lambda function
D. Add a ValidateService hook to the AppSpec file that inspects incoming traffic and rejects the payload if dependent services, such as the database, are not yet ready
Correct Answer: A
Reason: Add a BeforeAllowTraffic hook:
The BeforeAllowTraffic lifecycle hook is executed before traffic is routed to the newly deployed Lambda function. This is the appropriate place to verify that all dependencies (such as database changes) are ready before the new Lambda function starts handling requests. By adding logic here to wait for database propagation, you ensure that the function doesn't start receiving traffic until the database is fully updated.

Incorrect Options:
B. Add an AfterAllowTraffic hook:
The AfterAllowTraffic hook is executed after traffic has already been routed to the new Lambda version. Using this hook would not prevent the Lambda function from receiving requests before the database changes have propagated, thus it would not solve the issue of intermittent failures right after deployment.
C. Add a BeforeInstall hook:
The BeforeInstall hook runs before the new version of the Lambda function is installed. While this could delay the installation of the function, it doesn't guarantee that the database is ready before traffic starts flowing to the function.

Q12) A business has an application that consists of five independent AWS Lambda functions.
The DevOps Engineer has built a CI/CD pipeline using AWS CodePipeline and AWS CodeBuild that builds, tests, packages, and deploys each Lambda function in sequence. The pipeline uses an Amazon CloudWatch Events rule to ensure the pipeline execution starts as quickly as possible after a change is made to the application source code.
After working with the pipeline for a few months, the DevOps Engineer has noticed the pipeline takes too long to complete. What should the DevOps Engineer implement to BEST improve the speed of the pipeline?
A. Modify the CodeBuild projects within the pipeline to use a compute type with more available network throughput.
B. Create a custom CodeBuild execution environment that includes a symmetric multiprocessing configuration to run the builds in parallel.
C. Modify the CodePipeline configuration to execute actions for each Lambda function in parallel by specifying the same runOrder.
D. Modify each CodeBuild project to run within a VPC and use dedicated instances to increase throughput.
Correct Answer: Option C

Reason: Modify the CodePipeline configuration to execute actions for each Lambda function in parallel by specifying the same runOrder:
This approach allows the five Lambda functions to be built, tested, and deployed simultaneously instead of in sequence. Running the pipeline actions in parallel significantly reduces the total execution time of the pipeline because multiple functions are processed at the same time. By adjusting the runOrder setting in CodePipeline, you can specify which actions can occur concurrently.

Incorrect Options:
A. Modify the CodeBuild projects to use a compute type with more network throughput:
While increasing the compute capacity of CodeBuild can speed up build times in some cases, it won't solve the issue of running the builds in sequence. The bottleneck here is the sequential nature of the builds, not necessarily network or compute limitations.
B. Create a custom CodeBuild environment with symmetric multiprocessing:
CodeBuild doesn't support symmetric multiprocessing (SMP), and even if it did, this wouldn't necessarily improve the speed for independent Lambda functions. Each Lambda function is independent, and parallel execution via CodePipeline is a more appropriate solution than trying to process multiple builds in a single instance.
D. Modify each CodeBuild project to run within a VPC using dedicated instances:
Running CodeBuild within a VPC or using dedicated instances adds unnecessary complexity and is unlikely to improve performance in this case. The primary issue is the sequential execution of the Lambda builds, which can be addressed more effectively by parallelizing the builds in CodePipeline.

EXAM QUESTIONS FOR AWS SECURITY SPECIALITY (SCS-C02)

Q1) A company hosts a popular web application that connects to an Amazon RDS MySQL DB instance running in a private VPC subnet created with default Network ACL settings. The IT Security department has a suspicion that a DoS attack is coming from a suspected IP. How can you protect the subnets from this attack?
A. Change the Inbound Security Groups to deny access from the suspecting IP.

B. Change the Outbound Security Groups to deny access from the suspecting IP.
C. Change the Inbound NACL to deny access from the suspecting IP.
D. Change the Outbound NACL to deny access from the suspecting IP.
Correct Answer: Option C
Reason: Option C is CORRECT because restricting the inbound NACL is the solution to deny access for the suspecting IP address. A network access control list (Network ACL) is an optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets.

Q2) A company wants to have a secure way of generating, storing, and managing cryptographic keys, but they want to have exclusive access to the management of the keys. Which of the following can be used for this purpose?
A. Use KMS and the normal KMS encryption keys.
B. Use KMS and use an external key material.
C. Use S3 Server Side encryption.
D. Use Cloud HSM.
Correct Answer: Option D
Reason: CloudHSM allows you to securely generate, store, and manage cryptographic keys used for data encryption in a way that keys are accessible and managed only by you.

Q3) Your company has an EC2 Instance that is hosted in an AWS VPC. There is a requirement to ensure that log files from the EC2 Instance are stored in a secure manner. The access should be limited to the log files. How can this be accomplished? Choose 2 answers from the options given below. Each answer forms part of the solution.
A. Stream the log files to a separate CloudTrail trail.
B. Stream the log files to a separate Cloudwatch Log group.
C. Create an IAM policy that gives the desired level of access to the CloudTrail trail.
D. Create an IAM policy that gives the desired level of access to the Cloudwatch Log group.
Correct Answer: Option B and D
Reason:
Option B is CORRECT because you can create a separate log group and send all logs from the EC2 Instance to that group.
Option D is CORRECT because you need to create and restrict access to CloudWatch log groups using IAM policies.

Q4) You have an EC2 Instance in a private subnet that needs to access the KMS service privately within the AWS network. Which of the following methods can help to fulfil this requirement, keeping security in perspective?
A. Use a VPC endpoint.
B. Attach an Internet gateway to the subnet.
C. Attach a VPN connection to the VPC.
D. Use VPC Peering.
Correct Answer: Option A

Reason: When you use the VPC endpoint, communication between your VPC (EC2 within that) and AWS KMS is conducted entirely within the AWS network.

Q5) You have a website that is sitting behind AWS Cloudfront. You need to protect the website against threats such as SQL injection and Cross-site scripting attacks. Which of the following services can help in such a scenario?
A. AWS Trusted Advisor
B. AWS WAF
C. AWS Inspector
D. AWS Config
Correct Answer: Option B
Reason: AWS WAF allows you to create rules that can help to protect against common web exploits like SQL injection and cross-site scripting.

Q6) Due to compliance requirements, a company must rotate encryption keys every year. An AWS KMS key was created using imported key material. A security engineer needs a process to rotate the KMS key. Which key rotation process is MOST efficient?
A. Create a new KMS key and update the existing Key Alias to point to the new KMS key.

B. Upload new key material into the existing KMS key.
C. Select the option to enable automatic rotation for the existing KMS key.
D. Create a new KMS key and change the application to point to the new KMS key.
Correct Answer: Option A

AWS KMS keys with imported key material are not eligible for automatic rotation. Therefore, a new KMS key must be created with new key material. An Alias is a friendly name for an AWS KMS key and can be updated to point to a different KMS key. This is the most efficient method as it means any applications configured to use the Alias do not need to be updated.

Q7) A company is implementing an application that will use AWS KMS encryption keys. The company plans to create customer managed KMS keys within KMS and will not import any key material. The encryption keys should be rotated every 12 months. Which solution will meet these requirements?
A. Use AWS managed KMS keys instead so that AWS will rotate the keys automatically.
B. Schedule an AWS Lambda function to rotate the backing key of each KMS key.
C. Enable the option to automatically rotate each KMS key every year.
D. Change the customer managed KMS key policy to enable automatic key rotation.
Correct Answer: Option C
Reason: When you enable automatic key rotation for a customer managed key, AWS KMS generates new cryptographic material for the KMS key every year. You can enable or disable automatic key rotation for customer managed KMS keys at any time. You cannot modify the rotation configuration for AWS managed KMS keys which are automatically rotated every year.
Key rotation changes only the KMS key's key material, which is the cryptographic material that is used in encryption operations. The KMS key is the same logical resource, regardless of whether or how many times its key material changes. The properties of the KMS key do not change.

Incorrect Options:
- Use AWS managed KMS keys instead so that AWS will rotate the keys automatically: You cannot modify the rotation configuration for AWS managed KMS keys which are automatically rotated every year.
- Schedule an AWS Lambda function to rotate the backing key of each KMS keys: Customer managed KMS keys are being used so there is no need to create a custom solution as you can simply enable automatic key rotation.
- Change the customer managed KMS key policy to enable automatic key rotation: You do not enable key rotation through a key policy. Key policies are used to grant permissions to use a KMS key.

Q8) A security engineer has deployed a virtual security appliance inline. The virtual security appliance will be used to inspect traffic that is forwarded between subnets. What configuration is necessary to allow the virtual security appliance to route the traffic?
A. Configure the security appliance's elastic network interfaces for promiscuous mode and attach elastic IP addresses.
B. Place the security appliance in the public subnet with the internet gateway and route traffic to the appliance.
C. Attach an elastic network adapter (ENA), install the ENA module, and enable ENA support.
D. Disable the Network Source/Destination check on the security appliance's elastic network interface.
Correct Answer: Option D
Reason: Disable the Network Source/Destination check on the security appliance's elastic network interface.
By default, EC2 instances have source/destination checks enabled, which means the instance must be the source or destination of traffic it receives. Since a virtual security appliance needs to route and inspect traffic between other instances, it should not have this check enabled. Disabling the source/destination check allows the instance to act as a router or gateway, forwarding traffic to other destinations

Incorrect Options:
- Configure the security appliance's elastic network interfaces for promiscuous mode and attach elastic IP addresses:

Promiscuous mode is not a feature available or necessary on AWS Elastic Network Interfaces (ENIs). Attaching an elastic IP is for public-facing access and does not relate to routing traffic between subnets.
- Place the security appliance in the public subnet with the internet gateway and route traffic to the appliance:

Placing the security appliance in a public subnet with an internet gateway only makes it publicly accessible. This does not inherently allow the appliance to inspect and route traffic between private subnets.
- Attach an elastic network adapter (ENA), install the ENA module, and enable ENA support:

Elastic Network Adapters (ENAs) are used for high-speed networking but are not related to routing traffic or inspecting traffic between subnets. Enabling ENA support enhances network performance, but it does not enable the functionality of routing or inspection.

Q9) A developer recently left a company, and the company wants to ensure that any code the developer wrote cannot be deployed to AWS Lambda functions. The company uses AWS Signer for all Lambda functions.
Which solution will meet this requirement?
A. Revoke all versions of the signing profile assigned to the developer.

B. Use Amazon CodeGuru to review the code before it is deployed.
C. Remove all IAM permissions that grant access to AWS Signer.
D. Change the AWS KMS key that is used to encrypt the source code.
Correct Answer: Option A
Reason: Revoke all versions of the signing profile assigned to the developer - AWS Signer is used to sign code artifacts before they are deployed to AWS services like Lambda. By revoking the signing profile versions that the developer used, you prevent any future deployment of code that was signed using those profiles. This ensures that the developer's code cannot be trusted or deployed, even if it exists within the codebase.

Incorrect Options:

- Use Amazon CodeGuru to review the code before it is deployed:

Amazon CodeGuru is a tool for automated code reviews and performance profiling, not for code signing or deployment restrictions. It cannot prevent code from being deployed based on the author.

- Remove all IAM permissions that grant access to AWS Signer:

Removing IAM permissions would prevent future code signing but would not affect existing signed code. If the code was already signed with a valid profile, removing IAM permissions does not invalidate the existing signature.

- Change the AWS KMS key that is used to encrypt the source code:

Changing the AWS KMS key affects the encryption of the code but not the code signing. AWS Signer uses signing profiles to ensure the integrity and authenticity of the code, independent of encryption keys used for source code.

Q10) An attack left several Amazon EC2 Windows instances unresponsive. A security engineer has been asked to collect any memory dumps that may exist on the EC2 instances attached to Amazon EBS volumes. How should the security collect memory dumps for forensic analysis?
A. Reboot the EC2 Windows Server, enter safe mode, and select memory dump.
B. Run the EC2Rescue CLI using the /online mode and specify the instance ID.
C. Run the EC2Rescue CLI using the /offline mode and specify the device ID
D. Install the SSM agent and stream log data to Amazon CloudWatch Logs.
Correct Answer: Option C
Reason: EC2Rescue is a powerful tool for troubleshooting and repairing unresponsive EC2 instances. Using the /offline mode, you can run EC2Rescue on an instance by attaching the root EBS volume of the affected instance to a healthy instance. This approach allows you to access and extract memory dumps from the disk without booting the compromised instance, ensuring that the integrity of the forensic data is maintained.

Incorrect Options:

- Reboot the EC2 Windows Server, enter safe mode, and select memory dump:

Rebooting the server could cause the loss of volatile data such as memory dumps, and entering safe mode would alter the system state, potentially compromising the integrity of forensic evidence. This method is not a recommended forensic procedure.

- Run the EC2Rescue CLI using the /online mode and specify the instance ID:

/online mode is used for troubleshooting live, responsive instances. Since the instances are unresponsive, this mode cannot be used to gather the necessary memory dumps

Q11) A company has launched a web application running on port 80 on Amazon EC2 instances. The instances have been launched in a private subnet. An Application Load Balancer (ALB) is configured in front of the instances with an HTTP listener. The instances are assigned to a security group named WebAppSG and the ALB is assigned to a security group named ALB-SG. The security team requires that the security group rules are locked down according to best practice. What rules should be configured in the security groups? (Select THREE.)
A. An outbound rule in ALB-SG allows port 80 to WebAppSG.
B. An inbound rule in WebAppSG allows port 80 from source ALB- SG.
C. An outbound rule in ALB-SG allows ports 1024-65535 to destination 0.0.0.0/0.
D. An inbound rule in ALB-SG allows port 80 from WebAppSG.
E. An outbound rule in WebAppSG allows ports 1024-65535 to destination ALB-SG.
F. An inbound rule in ALB-SG allowing port 80 from source 0.0.0.0/0.
Correct answer: Option A,B,F

Q12) A company uses AWS across multiple Regions. A security audit highlighted some issues that must be addressed. The company must track all configuration changes affecting AWS resources and have detailed records of who has accessed the AWS environment. The data should include information such as which user has logged in and which API calls they made? What actions should be taken to meet these requirements?

A. Use Amazon CloudWatch to track configuration changes and AWS Config to record API calls and track access patterns in the AWS Cloud.
B. Use Amazon Macie to track configuration changes and Amazon CloudTrail to record API calls and track access patterns in the AWS Cloud.
C. Use AWS Config to track configuration changes and Amazon EventBridge to record API calls and track access patterns in the AWS Cloud.
D. Use AWS Config to track configuration changes and AWS CloudTrail to record API calls and track access patterns in the AWS Cloud.
Correct Answer: Option D
Reason: WS Config provides a detailed view of the configuration of AWS resources in your account, including changes over time. It tracks configuration changes, making it suitable for monitoring and compliance.
AWS CloudTrail logs all API calls made within your AWS environment. It provides records of user activity, including login information and API call history, which is essential for auditing and tracking access patterns.

Incorrect Options:

- Use Amazon CloudWatch to track configuration changes and AWS Config to record API calls and track access patterns in the AWS Cloud - Amazon CloudWatch is primarily used for monitoring and alerting on metrics. It does not track configuration changes or record API calls. AWS Config is designed to track configuration changes, not record API calls.
- Use Amazon Macie to track configuration changes and Amazon CloudTrail to record API calls and track access patterns in the AWS Cloud - Amazon Macie is used for data security and data privacy by identifying sensitive data in your S3 buckets. It does not track configuration changes. While CloudTrail can record API calls, Macie does not offer the required functionality to track configuration changes.
- Use AWS Config to track configuration changes and Amazon EventBridge to record API calls and track access patterns in the AWS Cloud - Amazon EventBridge is used for building event-driven architectures, not for recording API calls or tracking access patterns. AWS CloudTrail is the correct service to use for recording API calls and tracking access patterns.